



The Manasseh Hill Country Survey

*- Volume 4 -
From Nahal Bezeq
to the Sartaba*

Adam Zertal ז"ל
and Shay Bar

The Manasseh Hill Country Survey

Culture and History of the Ancient Near East

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Volume IV: From Nahal Bezeq to the Sartaba

By

Adam Zertal ז"ל

Shay Bar



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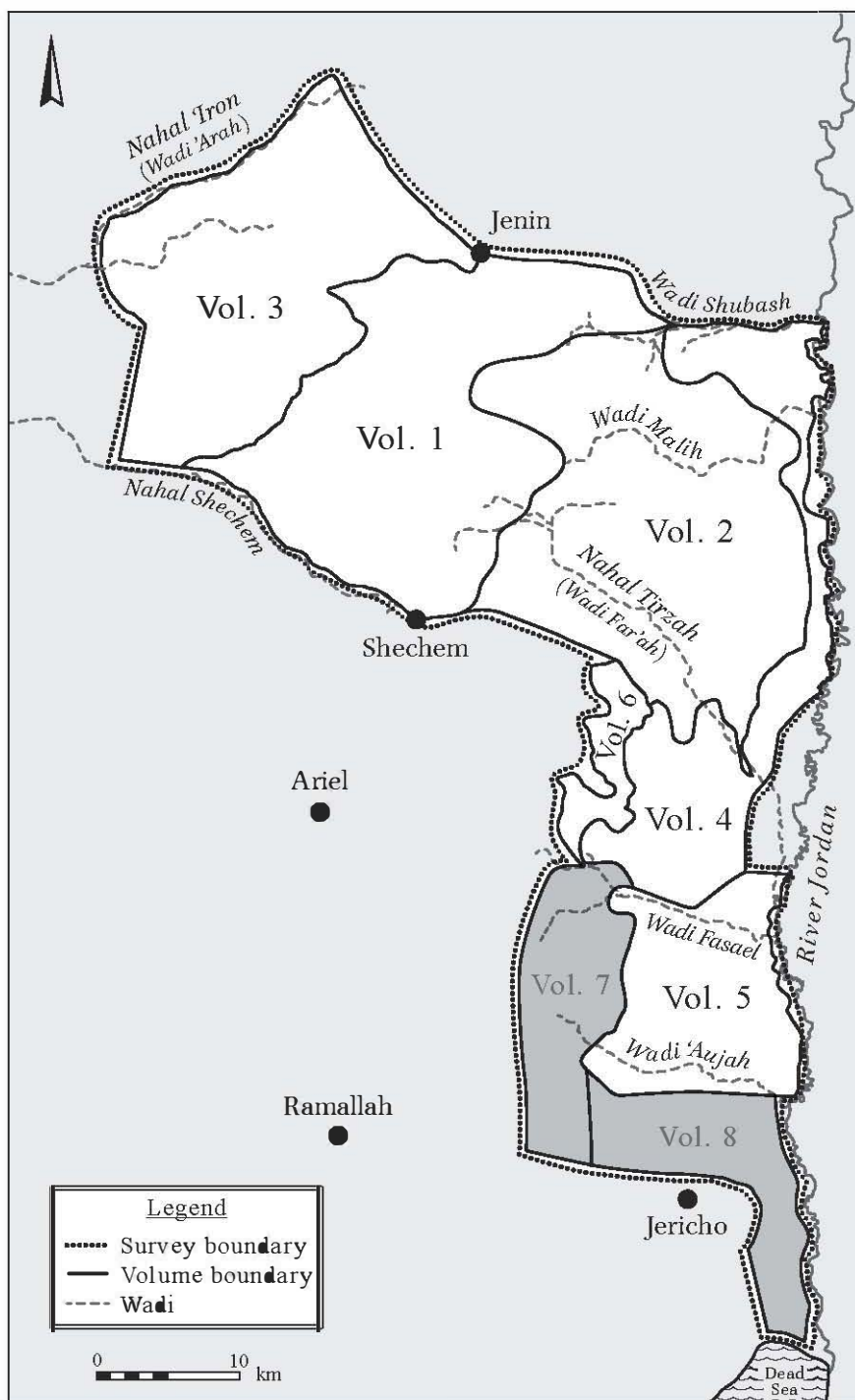
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1. The Manasseh Hill Country Survey team, 2004. Standing from right to left: Moshe Einav, Michal Oren, Daphna Ulman, Maya Nissan, Ya'el Ledder, Adam Zertal, Shraga Hashman, Haim Winter, Murad Tabber, Dror Ben-Yosef, and Amit Shadman. Kneeling: Ron Be'eri, Oren Cohen, and David Eitam. Missing: Gil Cooper and Haim Cohen (photo: S. Bar).



2. The Manasseh Hill Country Survey: Publication volumes.

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ABBREVIATIONS

a.s.a. - above surrounding area	MA - Middle Ages, Medieval period
a.s.l. - above sea level	MB, MBA - Middle Bronze Age
BCE, B. C. E. - Before Common Era	misc - miscellaneous
blk - black	mod - modern
brn - brown	No., nos - number, numbers
b.s.a. - below surrounding area	orn - ornamentation
b.s.l. - below sea level	Ott - Ottoman period
Byz - Byzantine period	PEF - Palestine Exploration Fund
CE, C. E. - Common Era	Pers - Persian period
Chal - Chalcolithic period	pk - pink
CP - cooking pot	rd - red
dec - decorated, decoration	Rom - Roman
dk - dark	sq. km - square kilometers
EB, EBA - Early Bronze Age	sq. m - square metres
EM - Early Moslem period	SWP - Survey of Western Palestine
EP, E. P. - Elevation point	Tal. - Talmud
ER - Early Roman period	T.F. - Threshing Floor
gr - grey	wh - white
ha - hectare	yel - yellow
Hel - Hellenistic period	Gen - Genesis
HM - holemouth (jar)	Chron - Chronicles
IA, Ir - Iron Age	Deut - Deuteronomy
IB, IBA - Intermediate Bronze Age	Jdt - Judith
ill. - illustration	Jer - Jeremiah
Kh. - Khirbeh (t)	Josh - Joshua
km - kilometer	Judg - Judges
LB, LBA - Late Bronze Age	Kgs - Kings
LR - Late Roman period	Macc - Maccabees
lt - light	Numb - Numbers
m - metre, metres	Sam - Samuel

PREFACE

This is Volume 4 of the Manasseh Hill Country Survey. The preceding three volumes have been published and accepted as an important contribution to the research of Samaria and the Land of Israel (Zertal 2004; 2008; Zertal and Mirkam 2016). This could be ascribed to the vast area surveyed (about 1500 sq. km in the preceding volumes, plus 250 sq. km in this one), the detailed data concerning each site, the new regions previously unexplored, and the methodology of the survey. All these offer the reader and researcher a source of information and fresh insights.

Volume 4 is not an organic unit. It is a mosaic of four landscape units, complementing Volume 2, and documenting the strips of land along the western side of the Jordan Valley. These strips are called *Ghor* and *Zor* in Arabic. The *Ghor* is the upper and wider western part of the valley, while the *Zor* is the lower strip closer to the River Jordan. The *Ghor* and *Zor* were not included in Volume 2 because of the differences between them and the adjacent desert fringes to their west, as explained below.

The first three landscape units of this volume: the southern Beit She'an Valley, the narrow Jordan Valley and the Nahal Tirzah floodplain, are linked by their relationship to the Syrian-African Rift and the River Jordan. The fourth unit: the Sartaba range and its surroundings, is exceptional; we have chosen to integrate it in this volume because of its relationship to the *Ghor* strip on one hand, and its relevance to the fringes of the adjacent desert on the other.

However, there are differences between the first three units. The first (Landscape Unit 27) belongs to the Beit She'an Valley, excelling in natural and agricultural richness and water sources. As a result, many settlements were founded in the area, making it one of the most productive in the land. This is also the link between the northern Beit She'an Valley and the narrow *Ghor* to its south. Unit 27 still belongs to the 'land of tells' in the north, but in comparison to the Beit She'an Valley itself there are fewer and smaller tells. This unit is also divided into sub-districts: for instance, the *Ghor* strip next to the River Jordan differs from the hill district in the vicinity of the village of Bardaleh, and they present different archaeological and historical data.

South of Landscape Unit 27 stretches the narrow strip of the Jordan Valley (Landscape Unit 28), which is about 24 km long and 1–2 km wide. It is an area cut by many ravines, almost entirely lacking in water sources, and was basically a link between the two other areas, although the importance of the road along it is uncertain. Most of the small settlements were found in the flood plains of the wadis descending from the desert fringes to the *Ghor*.

The third unit (Landscape Unit 29) is the Nahal Tirzah floodplain (Wadi

Far'ah). This is a large triangular area where the wadi approaches the Jordan. This area also connects the *Ghor* to Nahal Tirzah, has abundant water, and has been densely populated.

The nature of the fourth unit (Landscape Unit 30) is slightly different. The steep isolated Sartaba range constitutes a natural stronghold, and therefore a natural site for the fortress of Alexandrion which was erected on its summit. The characteristics of settlement here are unique: the ridge is surrounded by many small agricultural sites, mainly of the Iron Age II and Roman periods.

The fieldwork on this volume started in the early 1980s. At first the team worked in the districts of Mehola and the narrow Jordan Valley. The Sartaba range was intensively surveyed in the 90s. Additional work was done in 2014-15 as part of the final preparation of publication of the fully-revised English version.

In this volume we continue to employ the principles which have been developed through the years. These are described in detail in Volume 1, so are not repeated here. The fact that the region is open and accessible throughout the year enabled repeated visits to sites whenever required. This allowed accurate surveying and mapping and systematic gathering of finds. We have already stressed in previous volumes that measuring and drawing plans of the sites is an important part of an archaeological survey, and that it is difficult to understand a site without them. Indeed, most of the sites published in this volume with architectural remains were drawn and their plans are presented here. Introductions have been added to the descriptions of the sites – the principal part of the work – detailing natural data and summaries of the historical and archaeological processes affecting the region. These summaries express an opinion about the region surveyed and comparisons between it and the regions discussed in the previous volumes.

We have added important supplements, most of them not appearing in the original Hebrew version. The supplements include the important contributions by Haim Winter – The Flint Assemblages; Dror Ben-Yosef – Excavations at Bedhat esh-Sha'ab and Excavations at Yafit (3); Shay Bar – Excavations at Elevation Point -167 and Haim Shkolnik and Shay Bar – The Coin Finds. All of these present entirely new and illuminating material about the region and various aspects related to it.

Despite the geographical proximity of the territories covered in this volume and Volume 2, there is a difference: firstly, this volume is more diverse than the preceding one; and secondly, the investigation necessitated a longer spell of time in the field than those included in the other volumes.

In this volume we emphasize the archaeology of the Israelite settlement in the Iron Age, which influenced the traditions connected to the River Jordan and the Jordan Valley in the books of Joshua and Judges, and the controversy in research about the beginnings of the Israelite nation. The discovery of many

sites from that period, some of them complex (many south of the Sartaba), should lead to better understanding of the period. We hope that these discoveries will illuminate the economic and intrinsic social nature of the ancient Israelites.

Many new additions and changes have been made in the new fully-revised and extended English Volume 4, including:

1. Some of the site descriptions have been rewritten, based on additional visits and plans;
2. New sites have been added, based on new information, including extended additional surveys and salvage excavations in the area;
3. All the lists and indices have been updated;
4. New supplements have been added;
5. The pottery plates have been re-edited, with better coordination between the textual descriptions and the plates.

It is obvious that without our experienced team this major work could not have been accomplished. The devoted members of the Survey who worked in the area of this volume include the late Nivi Mirkam, Shraga Hashman, Gil Cooper, David Eitam, Moshe Einav, Haim Winter, Ron Be'eri, Dror Ben-Yosef, Murad Tabber, Haim Cohen, Shay Bar, Oren Cohen, Ya'el Ledder, Maya Nissan, Michal Oren, Amit Shadman, and Daphna Ulman. The plans were prepared by the Survey's team and drawn by Sapir Haad and Imri Zertal; field photography was by Moshe Einav, Ron Be'eri and Shraga Hashman; aerial photography was by Moshe Vineberg, Assaf Solomon and Moshe Einav.

It is a pleasant duty to thank everyone who assisted and sponsored the fieldwork and publication: the University of Haifa and our academic home - the Zinman Institute of Archaeology (with its leaders Mina Weinstein Evron and Michael Eisenberg); the Norwegian Organization for Israel and the Bible, headed by Gro and Bruno Wenske; the Pedagogica Foundation, represented by Abraham Ben-Naftali; the Hecht Foundation; and the Manasseh Regional Council. Additional assistance was given by Sapir Haad, who drew most of the plans and pottery; the Cooper family from Argaman, who opened their home to us; and Avraham Kopelman, who helped with the survey along the River Jordan and in the vicinity of Meḥola.

This volume was translated by Amatsia Halevi, and the English version was edited by John Tresman. Sapir Haad was in charge of the graphic design, and Oren Cohen proofread the text. The overall additional research in the field and in the laboratory, including the re-writing and complete organization of the English version, was done by Shay Bar.

Most importantly, thanks go to our families who endured our many days of absence from home.

Adam Zertal, May 2015

Addendum to the preface

Whilst finalizing the renewed field work at the sites described in this volume and starting preparation of this volume for publication, the Manasseh Hill Country Survey team and the Israeli archaeological community were shocked by the sudden death of the head of this long-lasting project – Professor Adam Zertal.

Adam was born in Kibbutz Ein Shemer on 3 December 1936, and was raised and educated there. He served as an officer in the Engineering Corps of the IDF. In 1959 he married Yehudit, and they had four children and 10 grandchildren. He served in various managerial duties, including economic director in his home kibbutz. At the same time he started to develop his interest in archaeology, and volunteered in many projects (Cave of the Pool, Masada, surveys in the Sinai Peninsula, the Emergency Survey, and others). From 1969 to 1972 he served in an Israeli mission to African countries on behalf of the Ministry for Foreign Affairs. He was an agricultural adviser and youth guide in the Central African Republic and Rwanda.

In the summer of 1973 Zertal decided to switch from economics and agriculture to archaeology, and enrolled at Tel Aviv University. He had not started his studies when the Yom Kippur War broke out in October 1973, and he was badly wounded in his hands, legs and one of his eyes. He underwent 12 operations, rehabilitation and skin transplants, while being hospitalized for almost a year. He was discharged from hospital in a wheelchair (with permanent 84% disability), and eventually switched to the crutches which became one of his hallmarks.

In 1978, Zertal started the project which occupied him for the rest of his life – the Manasseh Hill Country Survey. This continuous survey, diligently conducted by him for the past 38 years until his passing, is the longest, most comprehensive, and one of the most significant of archaeological surveys conducted in Israel. The methodology developed in the Survey (Zertal 2004, Chapters 1 and 2) was, and still is, a model for carrying out an archaeological survey on foot. The territory covered the area from Nahal Iron in the north to Nahal Shechem, and down to the northern shore of the Dead Sea in the south; and from the western slopes of the Samaria mountains in the west to the River Jordan in the east – 2500 sq. km, which previously had been practically unknown archaeologically. Over the years more than 3200 sites and archaeological findings were investigated and recorded, most of them previously unknown. About 2000 of these sites have been published in Hebrew in six volumes (four have so far been published in English).

During the years, Adam switched the hub of his academic activity to the

University of Haifa, where he was among the promoters of the new Department of Archaeology, and was also Head of the Department.

During the survey, four sites significant for the study of Israel were discovered and excavated: Mount Ebal, Arubboth/Narbata, El-Ahwat and the sandal-shaped enclosures (*Gilgalim*).

Mount Ebal – a double enclosure with structures within it; the most significant one identified by Zertal as an Early Iron Age sacrificial altar. The site, 2 km north-east of the summit of Mount Ebal, was discovered during the Survey, and excavated from 1982 to 1989 (Zertal 1986-1987; 2000). Two strata, with evidence in both of worship, have been found. The structure was established in the last



3. The late Adam Zertal, founder and director of the Manasseh Hill Country Survey from 1978 until 2015 (S. Bar).

quarter of the 13th century BCE, and lasted 50–60 years until being abandoned and concealed in an orderly manner. Zertal demonstrated the great similarity of the structure to the altar of the Second Temple in Jerusalem, as described in the Mishnah and other sources, and he proposed that this was the altar mentioned in the commandments of Moses (Deut, 27), where the Ceremony of the Blessing and the Curse (Josh, 8) took place. This proposal aroused fierce argument because it contradicted the proposition that these books were late, written towards the end of the First Temple period.

Arubboth/Narbata – Khirbet el-Hamam, located in north-western Samaria, was discovered during the 1978 survey, and was excavated from 1980 to 1984. According to Zertal this was Arubboth, capital of the Third Solomonic District, and Narbata, a Jewish town, destroyed by Cestius Gallus at the start of the Great Revolt (Josephus, War II: 495). In the excavation, previously unknown Roman siege works were discovered, comprising a circumvallation and a complex of military camps.

El-Ahwat – a large fortified city on the slope of the Katzir ridge, excavated between 1993 and 2000 by expeditions of the Universities of Haifa and Cagliari, Sardinia. The final report was published by Brill (Zertal 2012b). Two layers were unearthed: the later one was a 4th–5th century CE agricultural field. The earlier one was an Early Iron Age I fortified city, dated to 1220–1170 BCE, based on the ceramics and glyptic finds. Zertal proposed relating the unique architectonic characteristics exposed in the site to the Nuragic civilization, originating in Sardinia, and with one of the tribes of the Sea Peoples, the Sherden, who settled in the region at the beginning of the Iron Age. In addition Zertal proposed identifying the site with Biblical Harosheth Hagoyim – the city of Sisera.

The sandal-shaped enclosures ('Gilgalim') – this is a group of five sandal-shaped enclosures, discovered during the survey in the Jordan Valley and on the eastern slopes of Samaria. Three have been excavated, and dated to the Early Iron Age (see Appendices B and C). Zertal proposed that these are the 'Gilgalim' mentioned in the Bible (39 mentions of at least seven different sites). Lately he proposed that the sandal-shaped form was not accidental, testifying to the ideologically religious notion of possession of the land.

Zertal wrote many scientific books and articles, yet his great affections were his fieldwork and teaching. He passed away one week before the commencement of the 39th season of the survey, amidst the preparation for the new semester at the Kinneret Academic College and the Avshalom Institute, in both of which he had become a well-liked lecturer.

Zertal busied himself in numerous interests, in which he invested his endless energy. For many years he also published popular books in Hebrew, some of them presenting his discoveries to the general public (2000: *A Nation Born*;

2010: Sisera's Secret), and some fictional romances and short stories, unrelated to the archaeological sphere (1969: The Plain of the Wilderness; 2012: Period).

Adam Zertal will be remembered as one of the leading surveyors and researchers of the Land of Israel of recent generations, a true field-worker, and a principled and imaginative teacher and explorer. We, who have been following him in the field for many years, will remember him as a true friend and a passionate and devoted field archaeologist overcoming many obstacles in fulfilling his true love – revealing the secrets of his country's past.

Adam initiated this Survey in 1978, and was its director until the day he passed away planning the 2015-2016 field season. We will continue with the fieldwork and publication programmes in his memory.

Shay Bar, April 2016

PART ONE

INTRODUCTION

CHAPTER ONE

GEOGRAPHICAL AND SETTLEMENT DATA

A. History of Research and Historical Sources

For various reasons, already discussed (Zertal 2008: 11), modern research largely neglected Samaria and the Jordan Valley. Two types of researchers visited the region. The first group (Robinson, Guérin, the British Survey, Abel, Albright, Glueck and others) explored the area during the 19th and the beginning of the 20th centuries. The later group of archaeologists and geographers came in the second half of the 20th century (the members of the Emergency Survey, Mittmann, Tzori, Ilan and Damati, Mellaart and others).

Two aspects of the approach of the first researchers are worth noting. Firstly, their works were not field research as now interpreted (such projects are not very common even now), but their observations were broad-based and included information on the scenery, environment and local population. Their information was based upon visits to sites, most of them already known at the time, and on descriptions of the inhabitants' ways of life. As was common at the time, the findings were not systematically analyzed, and clear conclusions were not derived from them, but the information they contain is fairly plentiful. Secondly, the researchers tried to link most of the sites to the Holy Scriptures – the Old and New Testaments. In this way a sequence of reports was produced which had a fairly considerable descriptive value, especially because the terrain has greatly changed since their early visits.

One of the research pioneers was the American Edward Robinson, who visited the valleys of Jordan and Beit She'an in 1853. On his way from Shechem, he camped at Tell Mukehaz (Site 11), but did not report any ancient sites in the vicinity. His descriptions are only general, but his accounts of the way of life of the local inhabitants are especially interesting (Robinson 1970).

Victor Guérin, the noted French researcher, came to the region in April 1870. He focused on geographical-historical descriptions, and cited much from ancient sources. He included a great many names of locations and landscape descriptions, but the archaeological material is limited. His descriptions are fairly accurate and were written in a literary style.

Guérin ascended the Sartaba from the north, and on his way discovered the fort at Rujm es-Si'a (Site 86), which he called Kharbet Kerzeleia (Guérin 1969:

243). In the Jordan Plain east of the Sartaba he mentions tells which we could not find (Tell Oumm et-Theil and Tell Djouzelah – Guérin 1969: 233-234, 238). Guérin (1969: 235-238), mistakenly identified el-Makhruq (2) (Site 73) as Archelais (Ant. XVII, xiii: 1), and misnamed Tell eṣ-Ṣimadi (Site 84) as Tell Keraoua (Guérin 1969: 241).

After tours of upper Nahal Tirzah (Wadi Far'ah) Guérin turned north along the narrow Jordan Valley. He discussed Kh. Sakout (Khirbet es-Sakut - Site 32) in the south of the Beit She'an Valley and its possible identification with Biblical Sukkot (Judg 8: 4-8; I Kgs 7: 46). He, like Robinson before him, did not describe Tell el-Ḥammeh (Site 30), and continued northwards to Tell el-Ridhghah (Site 18).

The members of the British Survey, Claude Reignier Conder and Herbert Horatio Kitchener, explored the region from 1870 to 1874. The Jordan Valley and the desert fringes, in Volume 2 of their survey (Samaria), were included in sheets XII-XV of their map. For a description of the Jordan they cited the American naval Captain William Francis Lynch, who navigated the river and wrote about it and the valley as follows: "...the river, from its eccentric course, scarcely permitting a correct sketch of its topography to be taken. It curved and twisted north, south, east and west, turning, in the short space of half an hour, to every quarter of the compass – seeming as if desirous to prolong its luxuriant meanderings in the calm and silent valley, and reluctant to pour its sweet and sacred waters into the accursed bosom of the bitter sea. For hours in their swift descent the boats floated down in silence, the silence of the wilderness. Here and there were spots of solemn beauty. The numerous birds sang with a music strange and manifold; the willow branches were spread upon the stream like tresses, and creeping mosses and clambering weeds, with a multitude of white and silvery little flowers, looked out from among them..." (Conder and Kitchener 1882: 383).

Here also, the members of the British Survey did not deviate from their short descriptions. In a few cases they expanded a little about a certain site, for example Tell eṣ-Ṣimadi (Conder and Kitchener 1882: 395-396 [Kurawa el-Mas'udy]), Kurn Sartaba (Conder and Kitchener 1882: 396-401), Qa'un, Tell el-Ḥammeh and others.

The French researcher Felix-Marie Abel visited the region in February 1910, and made many significant observations (Abel 1913). His proficiency in the historical sources was noteworthy, and his attitude was much more scientific than that of his predecessors. When describing the springs in the vicinities of 'Ein ed-Deir and 'Ein Shamsiyeh, south of Kibbutz Tirat-Zvi, he was the first one to connect Tell el-Ridhghah to the battle of 23 January 635 between the Moslems and the Byzantines. While trying to clarify the 'Eynun traditions of the

New Testament, possibly occurring in this vicinity, he realized that there was confusion between the names of the various springs in the area. This confusion still exists, as a result of the frequent changes in the locations of the springs. On his way south he wrote about Tell el-Ḥammeh. He also mentioned the travellers preceding him and their opinions, and discussed the road ascending northward along the Jordan Valley.

During the 1920s, it was William Foxwell Albright who visited the region and devoted a brief discussion to the Bronze Age tells (Albright 1924-1925; 1925a). Amongst others, he identified the city of Ḥamath from the Egyptian sources in Tell el-Ḥammeh.

The Sartaba fortress excited great curiosity among explorers and researchers. The topographical dominance of the fortress and the historical significance of Alexandrion, described briefly by Josephus (see details in Site 173), encouraged researchers to visit the site; most of them climbing the ancient path ascending Wadi ej-Jeruzaliyeh from the north. It was Robinson who did not climb to the fortress, and only observed it from the distance. Based on the reports of researchers who examined the fortress, Robinson mentions an aqueduct apparently carrying water to the fortress from springs far afield. Among those who put forward new observations or ideas about the Sartaba we must mention Abel (who drew a schematic plan of the site); and Albright and Moulton, who noted the peculiar shape of the stones of the southern wall



4. The Sartaba range and fortress, view to the south-west, 2003 (M. Einav).

in the fortress, suggesting the possibility that the wall was built in the Middle Ages from ancient stones of earlier structures.

Few modern researchers have visited this region. Apart from the Emergency Survey of 1967, almost no archaeological-historical survey has been made in the region during modern times.

Some sites in Nahal Tirzah were surveyed by Glueck in 1943 when he visited western Palestine while he was working in Jordan (Glueck 1951).

In 1953 Mellaart (1962: 126-157) worked in the southern part of the Beit She'an Valley and along the River Jordan up to Jericho. Within the framework of his 'Archaeological Survey in the Yarmuk and Jordan Valley for the Point Four irrigation scheme' (mostly concentrated on the eastern side of the Jordan Valley), he surveyed 22 sites: among those discussed in this book were Tell Abu Sus, Khirbet es-Sakut, Tell Hammeh and others. He wrote very short descriptions of the sites, with no illustrative material. He was the only one to visit Kh. Sheikh Falikh (Site 38[1]), before mining operations prevented any further research there.

During the 1960s Mittmann (1970: 128-133), and Zobel (1966) worked in the southern part of the Beit She'an Valley. Within the framework of his 'Gil'ad Survey' (mostly concentrated in north-eastern Jordan), Mittmann surveyed ten sites, among them Tell el-Hilu, Khirbet es-Sakut and others. He also only wrote short descriptions of the sites, without illustrative material (plans, photographs or pottery drawings). Kappus (1966) and Knierim (1969) also surveyed several sites in Wadi Far'ah using the same methods.

In 1967, as part of the Israeli Emergency Survey, five sites were surveyed in the southern Beit She'an Valley by Gophna and Porath (1972: sites 39, 45, 46, 50 and 59). Their descriptions are brief, but the ceramic definitions were found to be generally accurate. Along the narrow Jordan valley they surveyed two sites (Gophna and Porath 1972: nos. 137 and 170), and three sites were noted in the Nahal Tirzah floodplain. Porath, who surveyed under the same framework, completed the Emergency Survey (his unpublished survey is called 'Samaria Survey B'), and added three sites to the corpus known at this time.

Bar-Adon's team, exploring the Judean Desert and the Jericho Valley in the framework of the Emergency Survey, discovered eight sites in the Nahal Tirzah floodplain (Bar-Adon 1972). The plans drawn by this team were of Rujm Abu-Muḥeir (Bar-Adon 1972: site 10), Khirbet el-Makhruq (Bar-Adon 1972: site 4) and the fortress at the Horn of Sartaba, of which only a preliminary sketch was prepared (Bar-Adon 1972: site 5). Two sites were recorded in the Sartaba ridge itself.

Nehemya Tzori surveyed the Beit She'an Valley during the 1960s (Tzori 1962). His second book was devoted to Issachar's allotted area, but he also penetrated into the southern Beit She'an Valley (Tzori 1977), where he described four more

sites, among them Tell el- Ridhghah, Tell el-Hammeh, and Tell Abu Sus (Tzori 1977: 53-56). Regarding Tell Abu Sus, Tzori repeated Mittmann and Zobel's suggested identification with Abel-Mehola. His work is noted mainly for its considerable pioneering value.

In the early 1970s, Ilan and Damati (1975) conducted a survey of Roman roads in the vicinity of the Sartaba.

In the 90s, Aharon Maier (1997) surveyed in the Beit She'an Valley as part of his PhD dissertation, concentrating on the pottery of the MBA II in the region. The same region was later partly surveyed by Achia Kohn-Tavor (2010), who revisited most of the known sites and added new data on them.

Numbers of sites visited in the main previous surveys are presented in Table 1.

Survey/ Unit no.	This Volume	Tzori 1962, 1977	Gophna and Porath 1972	Mittmann 1970	Glueck 1951	Bar-Adon 1972	Porath 1968 (Unpublished)
27	37	13	6	10			
28	40		2		2	1	
29	39		3		1	8	4
30	108				1	2	2
Total	224	13	11	10	4	11	6

Table 1: Numbers of sites visited in surveys.



5. The Nahal Tirzah floodplain, view to the south-east, 1998. In the foreground is the Sartaba range (A. Zertal).

A list of excavation projects follows:

Prehistoric excavations were conducted close to 'Urqan er-Rub (Hovers 1993) and at Huzuk Musa (Rosenberg et al. 2010).

There have been three excavations at the Middle Chalcolithic Tel Tzaf (Site 12): Gophna and Sadeh (1988-1989); major works at the site by Garfinkel (Garfinkel and Rowan 2005; Garfinkel et al. 2007; 2009; Ben-Shlomo et al. 2009), and renewed excavations by Rosenberg now in progress (Rosenberg et al. 2014).

Mellaart excavated a Chalcolithic site in the Jiftlik (Site 79) (Leonard 1992: 5-8).

Another Chalcolithic site, the Bezeq Canal, was excavated by Sadeh and Gophna (1991).

An Early Bronze Age Ib fortified site was discovered near Tell el-Ridhghah (Site 18), and its system of fortifications has been probed (Eisenberg 1996).

Excavations by Bar (Appendix D) were carried out at the Intermediate Bronze Age site of Elevation Point -167 (Site 192) in Wadi Ahmar.

The Iron Age sites of Bedhat esh-Sha'ab (Site 65) and Yafit (3) (Site 105), were excavated by Dror Ben-Yosef (Appendices B and C).

Several excavation seasons took place at Tell el-Hammeh under Cahill, Tarler and Lipovitz (Cahill and Tarler 1993).

There was a short excavation at Sartaba (Site 173) by Tsafrir and Magen (1993). The Sartaba water system, initially explored by the British Survey, was investigated again by Amit (1984). Meshel (1984; 1989) was the first to suggest the layout of the fortress siege works, and Einav (2014) added valuable information.

The Roman military camp close to Tell el-Ridhghah (Site 18) was partially excavated by Foerster and Gichon (HA 65-66 [1978]: 9).

The Staff Officer of Archaeology in Judea and Samaria conducted two salvage excavations in the region: Peleg (2002) excavated a Hellenistic structure near Maale Ephraim (Site 186[1]), and Magen (2004) excavated late Iron Age and Persian tumuli near Yafit (Site 102[1]).

B. THE GEOGRAPHY OF THE SOUTHERN JORDAN VALLEY AND THE SARTABA DISTRICT

1. *Boundaries*

The boundaries of the landscape units of Volume 4 complement those of Volume 2 of this Survey.

Throughout this volume the River Jordan is **the eastern boundary** of the area: The main features in the southern Jordan Valley (between Beit She'an and the Dead Sea) are the *Zor* (the lower strip of land close to the River Jordan) and the *Ghor*. A security fence was erected in the *Zor* in the early 1970s, stretching along the entire length of the border between Israel and Jordan, hampering and almost preventing the research of the strip next to the river. This strip is 500-700 m wide in the Beit She'an and Jordan valleys, and more than 3 km wide in the vicinity of the Nahal Tirzah reserve, in the east opposite the Sartaba. On several occasions we tried to survey some of the area and sites there, but we were only partially successful.

The northern boundary passes along Nahal Bezeq (Wadi Shubash). This stream divides the Beit She'an Valley into two: the larger part (about two-thirds) in the north, and the small one to the south. The two parts are very similar, but the Survey required a northern border.

The western boundary is the eastern boundary of Volume 2 of the Survey, and at the same time the topographical border between the Jordan Valley and the fringes of Samaria. At the north-west the border runs along the ridges of 'Iraq el-Hamam and Wadi Ha'shneh. From there, along the Bardaleh-Mehola road, it continues south-east to the ridge of Ra'us ej-Jibsin. From here the border turns south to the west of the Beit She'an-Jericho road (the Jordan Valley road). This road lies on the line between the desert fringes and the narrow strip of the flat Jordan Valley. After it reaches the Nahal Tirzah floodplain, the border turns west, and then south-east. It passes the southern foot of Ras el-Kharubeh to the narrow pass of the Shechem-Jiftlik road (Bab ed-Dayyq).

The Sartaba range protrudes westward further than the other three landscape units. This is a mountainous unit enclosed on all sides. Part of its border is the road south from the Hamra Junction (the 'Alon road'). In the north-west the border runs along the Alon road to Mehora, and from there south to Gittit and Maale Ephraim. From Maale Ephraim **the southern border** of the area in this volume descends eastward in Wadi Zamour and Wadi Ahmar until it reaches the Fasaal junction, from where it goes east to the River Jordan.

The total area of the four landscape units included in this volume is 250 sq. km.

2. *Geology and Geomorphology*

A. The southern Jordan Valley and Beit She'an Valley. The nature of the Jordan Valley, which is a part of the Syrian-African Rift, has great influence upon the settlement data contained in this volume. This rift, stretching for nearly 3000 km from eastern Turkey to East Africa, is a deep rupture in the crust of the Earth. The chronology and how it was formed are a subject for debate (Ilan 1973: 13-14). As to the time of its formation, there are at least three opinions. Several researchers are of the opinion that it was created by vertical tectonic movement. Others, who link it to the movements of continents, speak of horizontal fracturing. It is also possible that the Jordan Depression had split into a northern system (from Mount Hermon to Beit She'an) and southern one (from Beit She'an to the Arabah) during the Neogene. The considerable subsequent erosion partially filled up the riverbeds, including that of the Jordan, with alluvium. At the end of the Neogene or beginning of the Pleistocene, tectonic activity restarted, accompanied by erosion. This created the large alluvial fans – the Beit She'an Valley and the deltas (floodplains) of Nahal Tirzah and Wadi Ahmar. Lakes, which later dried up, developed over these fans during the Lower Pleistocene.

At the end of the Pliocene, some three million years ago, the Jordan Depression was connected to the Mediterranean and was flooded by sea-water. During



6. The chalky cliff of the *Ghor* and the *Zor* at its foot, view to the south, 2002 (M. Einav).

the following slow drying process, chalk, gypsum and various salts were deposited in the Beit She'an Valley. At the end of the Pleistocene the seas became disconnected, creating Lake Lisan, filling the whole Jordan Valley from the Sea of Galilee to Hatzeva in the Arabah. At the peak of the rainy period (about 25000 years BP) the lake's surface reached its highest level – about 180 m below sea level. Lake Lisan was salty and was connected to smaller secondary lakes. It is possible to trace this stage by the terraces of the lakeshore. The drying and retreat of the lake resulted in the creation of the *Ghor*. At the end of the process (some 15000 years BP), the drying of Lake Lisan left two residual lakes – the Sea of Galilee and the Dead Sea.

The River Jordan was created when water gushed out from the Sea of Galilee, and cut a chanel along the bottom of Lake Lisan. The river cut a meandering channel because of the very gentle gradient. The process began 12000 years ago in the Holocene and has continued until now. At the margins of Lake Lisan the streams continued to cut into the alluvium fans and the *Ghor*, creating today's typical transected landscape.

According to Nir (1962a: 102) the western upper level of the *Ghor* in the Beit She'an Valley is made of chalky tuff and chalky sediments of ancient lakes. The surface level of the *Ghor* appearing further east is made of Lisan marl – a lake sediment from the Mid-Quaternary. The *Zor* is made of alluvial soil brought by the river. South of it, in the vicinity of Nahal Bezeq, the moderate slope descends from the Gilboa to the Jordan, containing various alluvium fans of wadis.

B. The narrow Jordan Valley. From the vicinity of Meḥola southwards the Jordan Valley becomes very narrow. The region contains the eastern Samaria Mountains, descending steeply to the Jordan depression, as part of the Far'ah anticline (about this anticline and its formation, see a summary of an article by Meimran 1992, in Zertal 2008: 17-18). The fault system of eastern Samaria is orientated from north-west to south-east. These faults created the horsts (raised blocks of land) of Jebel Kebir, Jebel Ṭammun and Ras Jadir, which reach the Jordan depression. To the grabens (depressed blocks of land) belong (from south to north) Nahal Tirzah, the Buqe'ah and Wadi Maliḥ. Another part of this region is the *Ghor* and the *Zor*, discussed above.

C. The delta of Nahal Tirzah is a relatively young alluvium fan between the hard limestone of Ras el-Kharubeh and the Sartaba range. The fan is covered by a mixture of Mediterranean brown forest soils, rendzinas and terra rossa. This mixture is well suited for cultivation, and is relatively rich in organic matter.

D. The Sartaba range is a plateau of Eocene rock along the Jordan Valley, a southern section of the eastern Samaria range (Binyamini 1984: 8). The Sartaba is located on the lowest of the three tiers descending from the Samaria watershed to the Jordan Valley. The highest tier is situated on the Samaria watershed;

on the middle tier are located the villages of Kh. Yannun, 'Aqraba, Duma and Majdal Bani-Fadel; and the lowest tier (Mekhora, Gittit, Maale Ephraim) has fertile valleys. The Sartaba range rises to the east of the lower tier. It is a horst of soft rock, raised by a series of parallel geological faults. The considerable elevation differences (about 700 m – from 377 m a.s.l. at the top of the Sartaba cone down to 300 m b.s.l. in the Jordan Valley) caused especially severe erosion of the Sartaba slopes, and created the elevated formation of the horst. The Sartaba range is built of Eocene rock containing four formations: Zora, Nitzana, Horsha and Matred (Binyamini 1984: 8).

3. Climate, Water, Soil and Vegetation

Climate

The various districts have different climates. Beit She'an's climate is characterized by the transformation from the semi-arid climate in the north to a desertic (arid – Irano-Turanian) one in the south in the borderland of Samaria. The droughts here are relatively frequent, and therefore irrigated agriculture is customary (Nir 1962b). The average annual precipitation in the south of the valley is 288 mm, but the divergence from the average reaches 41%. The rainy season is lengthy, and the rain falls in brief and torrential showers (for example – two showers of 60 mm, lasting 35 minutes – Nir 1962a: 103). The temperatures are also noted for their extremes.

Further south, in the delta of Nahal Tirzah and the Sartaba range, steppe-like weather prevails, and the phytogeography is Irano-Turanian. In these districts the summer is long and hot with a steady wind regime along the valleys. The winter is moderate, with lower temperatures and little precipitation. The precipitation distribution through the year and between years is uneven (Katzanelson 1966: 51-62).

Water sources

There is a great difference between the Beit She'an Valley and other districts. In the south of the Valley (Landscape Unit 27) there are many water sources, most of them layer springs. We have counted 13 such springs: Mehazzim, Tzafot, 'Ein Ibrahim, 'Ein Muqehaz, 'Ein Bardaleh, 'Araq, 'Ein el-Bedha, 'Ein Buleibil, 'Ein Qurein, 'Ein Shams, Hammeh, Safsafa and 'Ein Sakut. The majority are on the eastern side of the region, and are or were the sources of streams, used mainly for irrigation.

The springs in the west (Qa'un, Bardaleh, 'Araq and Hammeh) emerge from above the Cenozoic aquifer in the seam between the mountains in the west and the floodplain of the Valley in the east. Settlements and/or multi-stratum tells developed close to most of them.

Knowledge of the influence of the many springs in the area is essential for understanding its history. Unlike other districts of Samaria, irrigated agriculture in the Beit She'an Valley evolved during very ancient times. Despite the fact that this issue has not been fully investigated, it can be assumed that small-scale irrigated agriculture already existed in the area in the Chalcolithic period. The relationship of the advanced Byzantine watering systems to those of preceding periods has not been clarified: was it irrigation with trenches, as in Egypt and Mesopotamia, or was it simply flooding irrigation? This issue requires additional research, but the link between the springs and the sites is clear.

The rather modest precipitation in the Jordan Valley, together with the inconsistency of the rains and the occurrence of floods, increased human dependence on reliable water sources (Bonne and Buydah 1977). In Landscape Unit 28, the narrow Jordan Valley, there are almost no flowing water sources. Along the 24 km of the narrow strip, there are only three small sources, at the edge of the *Ghor*: 'Ein Juneidiyeh, 'Ein Saleh and 'Ein Abu Sidreh. These are very small layer springs, with small-scale settlements nearby. In the Nahal Tirzah floodplain, Landscape Unit 29, there was an advanced channel irrigation system, and it seems that the whole area was irrigated. This system was noted by the Moslem traveller Yakut, who described cultivation of sugar cane in the Valley in Medieval times; and by travellers and researchers from the 19th and 20th centuries. The Ottoman (or earlier) aqueduct in Sheikh Mazar (Site



7. 'Ein Hāfireh spring south-east of the Sartaba, 1999 (R. Be'eri).

82) could have irrigated the entire western upper part of the delta, and it seems that similar aqueducts existed to the north of the riverbed.

The use of water from the Jordan for human consumption over the ages raises a question. The bed of the Jordan in the *Zor* is below the level of the surrounding area, and it seems that aqueducts did not exist along its course. The frequent changes in the course of the Jordan, due to its very slight slope, also made the use of aqueducts impractical. The presence and area of agricultural land along the Jordan during ancient times were not investigated because of the uncertain state of security and the minefields. The maps and the few earlier surveys in the region indicate very few tells in the *Zor* itself. Such tells have been found only at the south of the Beit She'an Valley (Tell Jima'in, Tel Tzaf and Tell Abu Sus). Further south, only Tell Abu Sidreh, located in the *Zor* close to the Jordan, is noteworthy. The eastern side of the Jordan Valley, especially the valley of Sukkot, is much richer in water sources and soil than the western side, hence the scarcity of tells on the western side and their abundance on the eastern side. Habitation was always concentrated on the eastern side, helped by the absence of political frontiers between east and west.

The mountainous land-block of the Sartaba is surrounded by several springs with medium or low flows, mostly on its eastern and western sides. In the east is 'Ein Jeruzaliyeh; and in the west 'Ein Abu Daraj, Hafireh and 'Ein el-Mana. Most of the approach roads to the Sartaba developed near these springs, beginning in the Iron Age.

The soil

The types of soil are categorised according to the landscape units (for general review see Dan 1977):

1. The south of the Beit She'an Valley is covered by a mixture of alluvial soil with remnants of Holocene lake soils (Ilan 1973: 48). The area of the valley next to the Jordan is covered with sedimentary Lisan marl, rich in gypsum and salt. This soil is poor in organic matter, and is not very fertile. The agricultural wealth of the district relies more on spring water than soil fertility.
2. The soils of the narrow Jordan Valley and Nahal Tirzah's flood plain are a mixture of rendzinas, washed-out desert and colluvial-alluvial soils.

In the high *Ghor* and in the *Zor* there are concentrations of gypsum and salt in the Lisan marls, causing the formation of saline flats (*sabkhas*) in the Nahal Tirzah floodplain. Brown forest soils from the desert fringes of Samaria are washed out and accumulate in the alluvial fans. These are fairly fertile soils, in which wheat and other seasonal crops are grown.

Date stones found in the Teleilat Ghassul excavations prove that date palms were already cultivated in the *Ghor* in the Chalcolithic period (Levy 1993: 509). The epithet 'The Valley of Jericho, City of Dates' (Deut 34: 3) shows that cultivation of date palms was common in the Iron Age. Vast groves of date palms and persimmon (*Diospyros kaki*) were grown in the Lisan marl soils of the Jordan Valley in the Roman period (War I, xviii: 5).

Vegetation

Zohary (1980: 186-190) divides Samaria into two phyto-geographical districts: Mediterranean and savannah-like (Irano-Turanian). Damin (1970) divides eastern Samaria into three longitudinal strips and four main plant communities. Based upon the work of Berliner (1978) and Sabah (1992), the landscape units of this volume may be classified as follows:

The Beit She'an Valley, the narrow strip of the Jordan Valley (Landscape Unit 28) and the alluvium fan of Nahal Tirzah belong to the tropical enclaves with Sudanese-Dicakan vegetation (Zohary 1955; 1980: 187). These are districts with high temperatures and flowing water, dominated by Jericho Balsam (*Balanites aegyptiaca*), Common Jujube (*Ziziphus spina-christi*) and Sodom Apple (*Calotropis procera*).

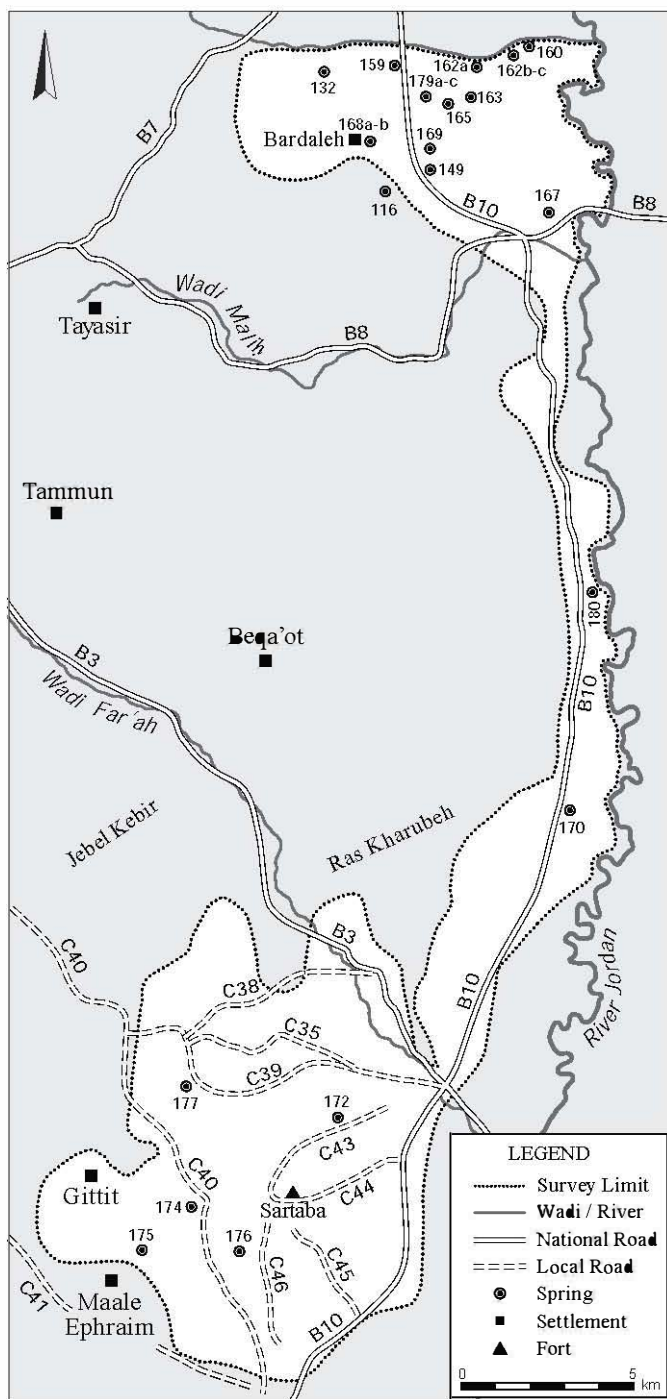
In the narrow Jordan Valley there is no flowing water and arable land is scarce; therefore the savannah and Sudano-Dickian vegetation grow together.

In contrast, in the Sartaba range a Mediterranean borderland floral community dominates, together with Irano-Turanian flora. The vegetation here is herbal without trees (Sabah 1992: 50). The sole plant to reach the dimensions of a small tree is the Jujube. Also found here are communities of White Broom (*Retama reatem*), *Artemisia Sieberi*, Sharp Varthemia (*Chilliadenus iphionoides*) and various species of Germander (*Teucrium* spp.).

4. Roads

4.1 Roman Roads

The good state of preservation of the Roman roads results from the sparse settlement in the desert fringes, the aridity and the infrequent human activity. This situation is similar to that in the Negev and the eastern Jordan Valley (for a general review see Roll 1976). With the increase of settlement, the advancement of agricultural development and fortification-building by the army in recent decades, many of the remains of Roman roads have been destroyed. Despite the good state of preservation, the Roman roads have been only superficially investigated. Conder and Kitchener first included them on their maps in 1870-1880. The roads were still visible during the British Survey, also by Glueck, and



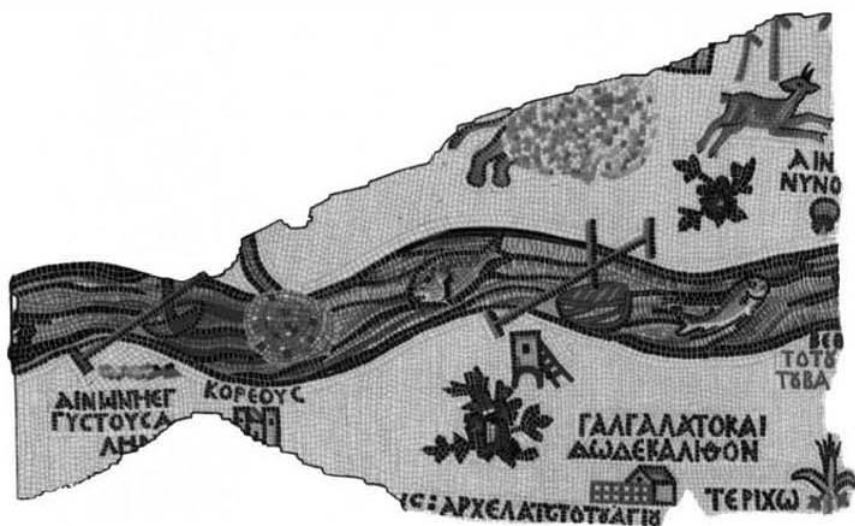
8. Map of the borders of Volume 4 and the main roads and springs, with their serial numbers.

even at the time of the Emergency Survey (1967). From 1973 to 1975 Ilan and Damati (1975) conducted a partial survey of Roman roads in the vicinity of the Sartaba. Some were recorded, and several sites near them were noted. Others (for example Dorsey 1991) summed up data without exploration in the field. In the case of Dorsey, the Iron Age sites found in the Emergency Survey were connected to each other by straight lines, creating a very illogical web of roads.

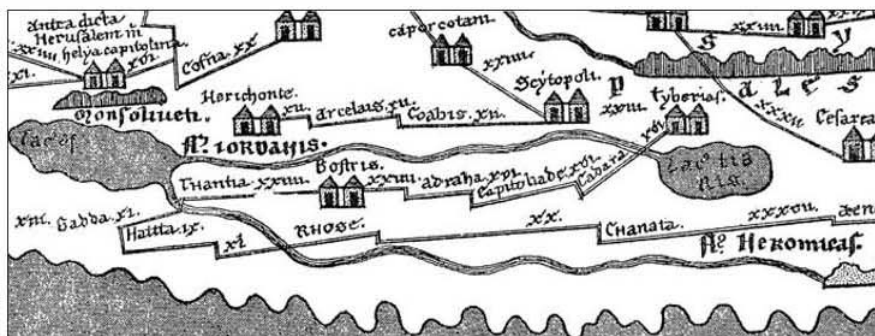
The roads:

1. The best known Roman road leads from Beit She'an to Jericho (B1● according to our terminology). Avi-Yonah (1963: 91), who relied on Thomsen (1917: 73), suggested that the road was paved in the time of Hadrian. Along it, north and south of Beit She'an, were found several milestones (Tzori 1962: sites 109-125): one from the reign of Caracalla, and the rest from that of Marcus Aurelius. In the area of our survey, south of Nahal Bezeq, some 8●● m south of Tirat Zvi, 11 fragments of milestones with remains of inscriptions have been found: one inscribed IMP C---/ANTONI/FE---/BRIT, and another -NINO---/A-O CKYOO-/VI (Tzori 1962: site 118). Mittmann (197●: 151) published a milestone (without inscription) found in a field, 1.5 km west of Khirbet es-Sakut (coordinates 1974/2●●2), where the tenth mile marker from Scythopolis was supposed to have been.

The road from Beit She'an to Jericho has been discussed several times (Mittmann 197●: 151; Zertal 1991). Several forts were found along it. It could be that the road appears in the Madaba Map (Avi-Yonah 1953: pl. 5), in which are shown the settlements of 'Einim near Shalem', 'Coreae', 'Phasaelis' and 'Archelais'.



g. The Madaba Map - the area of our survey (Avi-Yonah 1953: pl. 5).

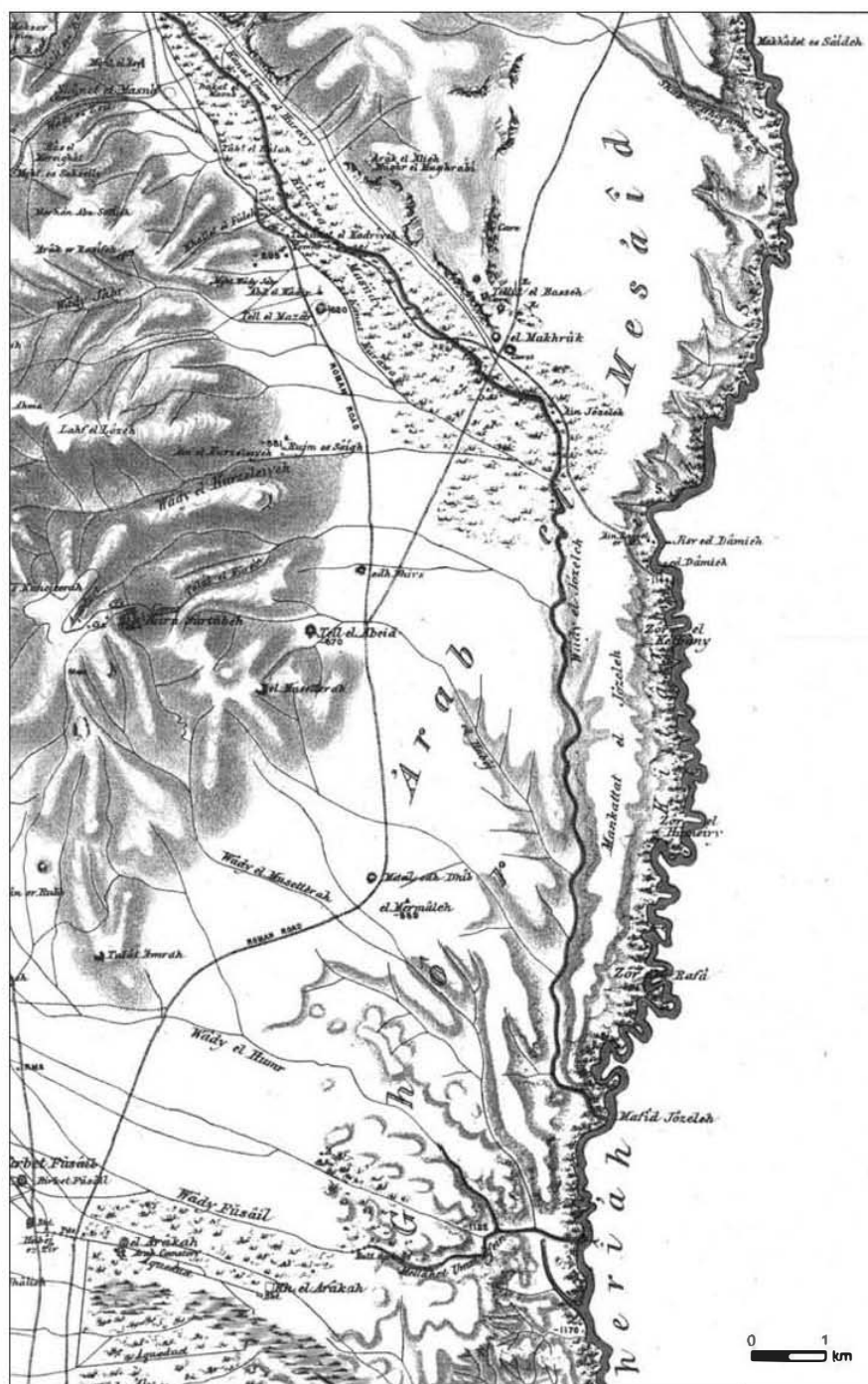


10. The Peutinger Map - the road from Scythopolis to Herichonte (Jericho) and the region of the Survey (Miller 1962: Segmentum X: 1-2).

In the Peutinger Map (*Tabula Peutingeriana*) the road is in three sections, with two stations (Ilan 1973: 149). Coabis is the first station, 12 Roman miles south of Beit She'an (Scythopolis). The next station is Archelais, 12 miles from Coabis. The third leg from Archelais to Jericho is also 12 miles. This means that the total distance is 36 Roman miles, about 53 km. All stations, apart from Coabis, have been identified with varying degrees of certainty. Zertal (1991: 13) has suggested identifying Coabis at 'Iraq Abu Hashish (A)' (Zertal 2008: site 248) or in Kh. Khiraf (Zertal 2008: site 266). The stations in the map seem to be fortified posts, and likewise have been found in both sites. The site at Kh. Fusayil was surveyed in 2000 (Zertal 2012a: site 34), and it was established that there also is a fort of the same type, thus somewhat strengthening the identification of the stations as forts.

In the map of the British Exploration Fund (sheet XV) the road from Beit She'an more or less follows the route of the existing road to the Valley Monument. Glueck found the remains of the road, close to Kh. es-Saleh, in 1943 (Glueck 1951: 420). The road is up to 6 m wide at this point. In the Emergency Survey, next to the Valley Monument, Bar-Adon (1972: site 14) traced a section of the road, and described it as follows "Remains of a Roman road, marked in the maps, on its sides two forts: one north of the road and the second at its south". Despite our search, some 35 years after the Emergency Survey, we have failed to find this road.

2. The road from Beit She'an (Scythopolis) to Shechem (Neapolis). This section of the road (B7 – according to our terminology – Zertal 2008: fig. 8) runs next to Tell Qa'un (Site 3), and from there to the entry to Wadi Shubash (Ilan 1973: 150). The road was published in the British Survey (Conder and Kitchen 1882), by Thomsen (1917: 70-72), Avi-Yonah (1963: 88) and others. It was also discussed in our Volume 2 (Zertal 2008: 28). Several milestones were found along it, close to Nahal Bezeq (Tzori 1962: site 113), and identified by us.



n. The SWP Map (sheet XV) including the Sartaba range and eastern Wadi Far'ah.

Within the bounds of our survey, Thomsen noted the following stones: no. 248 – south of Kh. Qa'un, with the inscription: IMP CAES IVLIO MAXIMINO PIO; no. 249 – the XXth mile from Neapolis, with the inscription: CAES C IVLIO UERO MAXIMINO P FEL AUG MANOCOMAX I MAXIMO NOBILISSIMO CAES FIL EIUS. Some of the milestones bear inscriptions from the reigns of Marcus Aurelius (162 C.E.) and Caracalla (211-217 C.E.). In the biography of the monk Sabas (Ilan 1973: 150) this road is mentioned as the 'King's Highway'. No remains of the road close to Qa'un were actually found.

3-5. The roads from Jericho to Shechem (Neapolis), have been noted as roads 1 and 2 in Ilan and Damati's survey (1975: map in p. 44), and C40 and C41 in our survey. From Jericho the road goes north in a straight line to Kh. Fusayil (Zertal 2012a: site 34). A fine description of this section appears in the SWP report: "The Roman road is very perfect near this ruin; it consists of three parallel lines of stones, about 1 foot square, forming the sides and central rib of the road, 9 feet apart, giving 18 feet for the width of the road. There seems to have been no foundation or drainage, but the central rib is higher than the sides, so that the road had a section of two inclined planes. The part between the ribs is filled in with a sort of cobble of stones of irregular size, covered with smaller metalling. The central rib consists of two lines of stones, the other wall, or curb-stone, of a single line each side" (Conder and Kitchener 1882: 392-393).

At Kh. Fusayil the road branches into two: one (our C40, Ilan and Damati's 2) ascending north-north-west, via the Wadi Ahmar valley, reaching the vicinity of 'Ein Hāfireh. Here it turns north-west towards Wadi Zamur and climbs to Neapolis via Sahel Afjeam, Wadi Kard and Ṭana et-Tahta. The leg ascending from Wadi Ahmar, over a narrow ridge to the site E.P. 150, deserves particular attention. Here, in the south-eastern part of the Sartaba range, the Roman road is well preserved. This section, measured and registered, received a special site number – 177 – in this volume. No additional remains have been discerned in the continuation of this section to the north-west. A very long stretch of this road has also been preserved to the east of the large site at Kh. Ṭana el-Foqa (Bar and Zertal 2016: 29-30).

The other branch (C41, Ilan and Damati's no. 1) ascends north-westwards from Kh. Fusayil over the ridge and south to Maale Ephraim, until just before the village of Majdal Bani-Fadel. The road passes close to Rujm Abu Mukheir (Bar-Adon 1972: site 10; Zertal 2012a: site 5); a section of which appears in Bar-Adon's round fort plan and is still visible. This branch ascends to 'Aqraba, and from there to Neapolis.

6-9. The roads from the Damia Bridge (the Jiftlik) to Shechem (Neapolis). This is the principal road (B3 in our terminology), by which Vespasian descended from Neapolis to Korai (War I, vi: 5), along Nahal Tirzah.

The three other roads recorded in the survey connect the Jordan Valley (the Jiftlik area) with Neapolis via Kh. Ṭana et-Tahta (Bar and Zertal 2016: 27-32). These roads are:

- Road C38, which ascends to the Es-Sawaneh plateau via Wadi ez-Zeith, connecting to C35 near Wadi edh-Dhba'ah' (3) (Site 131). The road's remains have not actually been found, but various clues, mainly the settlement model, hint at its existence.

- The second road (C35) ascends from the east to the 'Iraq er-Resifeh plateau. From there it runs along the northern crag of the ridge of the Sartaba, via the sites of Jebel el-Maḥjarah (5) (Site 139) and El Mureighat (E.P. 280) (Site 133). Then it reaches Wadi edh-Dhba'ah' (3) and connects to road C38, and afterwards to road C40. In these areas, sections of the road and sites built along it have been preserved. Most are forts containing fortified towers with adjoining courtyards.

- C35 and C38, joining C40. A section of this road has been preserved in the saddle near E.P. 236 (a small fort, Site 119), sloping sharply downwards to the Alon road. From here it ascends directly in Wadi Kabi to Kh. Ṭana et-Tahta.

- The next road is C39, which ascends Wadi Jaber westward to 'Ein Abu Daraj. A fine stretch of this road has been preserved next to the Lahmeh Complex (Site 155), and from here it continues west to Kh. Ṭana et-Tahta.

4.2 Other Roads

1. The road from Beit She'an to Jericho (B10 in our terminology) is not mentioned in the Bible. The district is part of the Biblical Arabah (Deut 3: 17; 11: 30, etc.), but the expression 'The Arabah highway' (II Sam 4: 7; Jer 39: 4) probably refers to the road from Jerusalem to Jericho and to roads on the eastern side of the Jordan. Aharoni (1979: 57-58 and map 3) mentions a road west of the Jordan parallel to the one on the eastern side; however, we failed to find any literary mention of it. Possibly, the pair of oval Iron Age II forts which have been discovered above the road on the east side of the Sartaba (Sites 98, 99), are connected to this road. It is difficult to interpret these forts without a context to the Beit She'an-Jericho road; it is interesting that Iron Age pottery was also found in Kh. Beiyudhat (Archelais) (Zertal 2012a: site 111), located further south along the road.

The main north-south longitudinal road probably passed along the eastern side of the Jordan. That side, abundant with water and agricultural land, was always more populated than the narrow arid western side (Mazar 1986; Aharoni 1979: 57-58). Due to the fact that the eastern side is much wider than the western one, the routes of the roads are more convenient.

Most of the Biblical roads mentioned in this volume are transverse

(East-West) routes, linking the central hill country with the cities of Shechem, Tirzah and Samaria to the Jordan Valley and westwards. Among them we note the following roads, in north to south order:

2. B7, from Shechem to Beit She'an via the Bezeq Pass. King Saul and his army went this way to come to the rescue of the inhabitants of Yavesh Gilead (I Sam 11: 8 ff; Zertal 2008: 28). This route is identical with the Neapolis-Scythopolis Roman road, descending from Shechem via the Tubas Valley to the Bezeq Pass, Wadi Ha'shneh and Beit She'an. This road, passing the complex of Qa'un sites, certainly contributed to their importance.

3. The other transverse road (B8 in our terminology – Zertal 2008: 28) descends eastwards from the Tubas Valley to Wadi Maliḥ. Within the bounds of Landscape Unit 27 (the southern part of the Beit She'an Valley) the road runs eastwards from the vicinity of Kh. esh-Shaqq along the bed of Wadi Maliḥ, crosses the present Valley road and reaches the River Jordan. At this point the two forts of Kh. esh-Shaqq are worth noting (sites 82, 83 in Zertal 2008), probably connected to this road. One is a fort consisting of three concentric circles (Zertal 2008: ill. 195) of the Ammonite type from Iron Age II. Identical forts appear by the Nahal Tirzah road and next to the Shechem–Jericho road. These three circular forts mark the locations where the roads begin their ascent from the Jordan Valley westward to the spine of the hill country and to the centres of rule of the northern Kingdom of Israel– Shechem and Samaria.

Zertal (1988; 1995), who dealt with their relation to the Ammonite forts of the Rujm al-Malfouf type, suggested identifying them with the Biblical “Migdaley Nozrim” (watchtowers). It is also likely that the sites of el-Marma (from later periods, Site 34) and Wadi Maliḥ (from Iron Age I, Site 35) served a similar purpose. It seems that the crossing of the Jordan used to be via the two fords on both sides of Tell Abu Sus (Site 33) – Maḥdeth Faddalla and Maḥdeth Abu Sus. There is no significance regarding the identification of Abel Meḥola with this tell because the Bible does not relate the city to the road (see discussion in Chapter 2).

4. The next transverse road is that of Nahal Tirzah (B3), identified by Zertal as the Biblical road ‘where the sun sets’ (Deut 11: 30; Zertal 2000: 240-259; 2008: 28). It may be that some sites in the Nahal Tirzah floodplain, specifically Tell eṣ-Ṣimadi (Site 84), were connected to this road, which crossed the Jordan at the Adam Crossing (Damia Bridge). In our opinion, Tell Eṣ-Ṣimadi served as a foundation of forts at various periods. This assumption is based upon the shape of the tell and its resemblance to the site at Tell esh-Sheikh Diyab near Fasael (Bar-Adon 1972: site 13; Zertal 2012a: site 23). Both tells are constructed as a foundation with steep sides designed to elevate the fort above the surrounding plain. To this group could be added the forts at El-Makhrūq (one is of the

circular Ammonite type – Yeivin 1992: 157-160; the other, rectangular-rounded, at the same location [our Site 73]) and the sites at Bab ed-Dayyq (Zertal 2008: site 197) and Sheikh Mazar (Site 82). The last one, in which the British Jiftlik police post is located, is derelict.

5. The last transverse road listed here descended from Shechem via 'Aqraba to Jericho, and it seems that parts of the boundaries between Manasseh and Ephraim passed along it (Josh 16: 6-8). Branches of this road run through Wadi Afjeam and Maale Ephraim, and next to it is another circular fort of the same type (Bar-Adon 1972: site 10; Yeivin 1992: 156-157). The discussion of this road is not included here.

6. Roads ascending to the Fortress of Alexandrion (Sartaba): The routes of the ancient paths ascending to the fortress have been marked according to our judgment as follows:

6a. Wadi Jeruzaliyeh road (C43) used to be an important ascent, probably the main one, to the fortress. The road leaves the Jordan plain and enters Wadi Jeruzaliyeh to meet the water system siphon. In some places the ancient path, including improvements made to it, can be seen. Rujm es-Si'a (Site 86) is located in the entrance at the east, probably to guard the road. Most researchers and visitors climbed this route until the paving of the present roads.

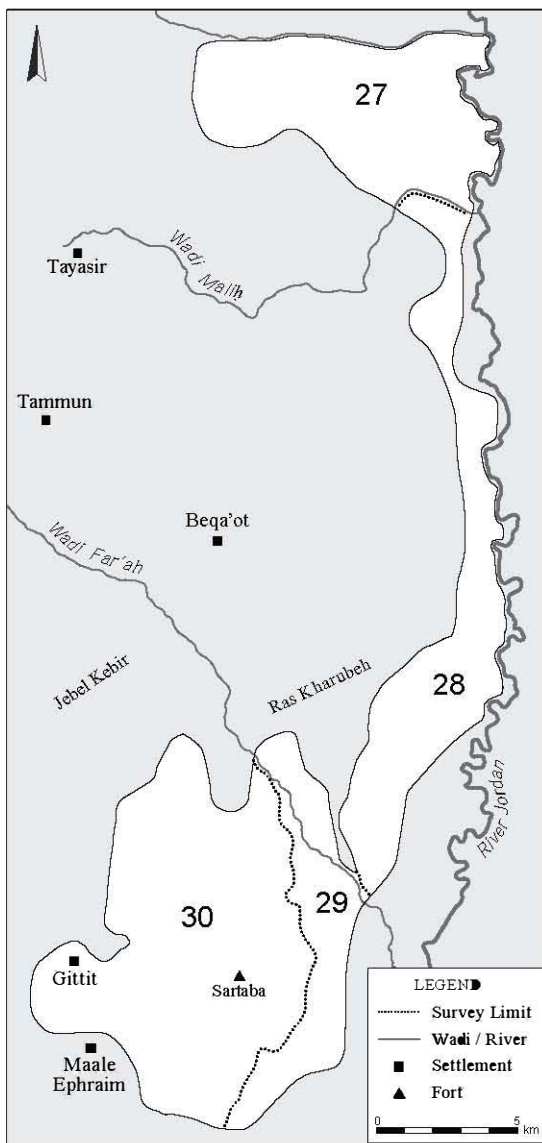
6b. The 'Urqa el-Mastarah path (C44): the path probably ascended to the saddle at coordinates 1948/1664 (the route of the new road), and from there climbed along the narrow spur eastward to the fortress. There is a modern path, most likely to have been used in ancient times, at the place. No trace of the ancient path was recorded, but the topography and the modern path suggest the possibility that it was also used in ancient times.

6c. Wadi Qurein road (C45): the road approached from the south-east and met the wadi at coordinates 1947/1641, north-west of Yafit. From there the road went by the convenient route of the wadi to reach the steep climb to the fortress. It should be noted that the climb from this spot is especially difficult, and we are not sure that this road actually reached the fortress. Meshel (1970: 105) was the first to note a set of stairs in the upper fortress possibly leading in this direction. The documented end of this path is a very unusual road built with superior masonry (Site 177).

6d. The ascent via Wadi 'Amrah (C46): It seems that the ascent was via the Yafit (3) valley – Site 105, and from there by a path northward, which still exists. This path conveniently exploits the plateau above Wadi 'Amrah. It reaches the quarry site of the Lower Sartaba – Site 172, and from there climbs to the fortress. At coordinates 1927/1660 rock-cut stairs have been discovered, probably belonging to this road (Site 177).

C. LANDSCAPE UNITS

This volume contains four landscape units. For defining the landscape units we have employed the principles crystallized when writing Volume 1 (Zertal 2004: 13-14): the variety of natural factors, clearly geographically bounded landscape units, consolidation of small areas into larger units for the sake of wider scope analysis and the historical data.



12. Map of landscape units.

Landscape Unit 27 – Southern Beit She'an Valley

This unit has natural boundaries on three sides: Nahal Bezeq to the north, the River Jordan to the east and Wadi Maliḥ to the south. The western border is a seam between the eastern fringes of Samaria (ʿIraq el-Ḥamam and the area of Bardaleh-Wadi Salman) and the plain of the southern Beit She'an Valley. The area of this landscape unit covers some 45 sq km: it is divided into two sub-districts with historical-archaeological significance.

A. The Bardaleh hills, of area about 13 sq. km, stretching from Nahal Bezeq to the village of Bardaleh, and from the bottom of ʿIraq el-Ḥamam to the Jordan Valley road. It is a land of low hills, convenient for cultivation and travel. The area is drained by the tributaries of Wadi Ha'shneh in the south and Wadi Shubash (Nahal Bezeq) in the north. There are several springs, the dominant ones being those of Qa'un. Around them evolved a rich multi-period settlement complex. In the south are two layer-springs next to the village of Bardaleh – ʿEin Bardaleh (168a) and ʿEin el-ʿIraq (168b): however, not many sites have been found in this area. A settlement complex has been discovered by the third spring, ʿEin Mukehaz (159), which is not as rich as the spring near Qa'un.

There is a considerable concentration of sites in the hills of Bardaleh. Small sites are scattered in the area between the three springs and on the hills, many of them from the IBA. The significant sites are those of Qa'un. This sub-district is crossed by two roads: B7, descending from Shechem to Beit She'an, and B10 (Beit She'an–Jericho). Both the roads influenced the nearby settlements.

B. The *Ghor* and the *Zor*, between the Valley road in the west and the River Jordan in the east, and between Nahal Bezeq in the north and Wadi Maliḥ in the south. The area is about 32 sq. km. The *Ghor* here is a flat plain sloping gently eastward. From the vicinity of ʿEin Mukehaz (140 m b.s.l.) it drops 100 m over 5 km, sloping 2% to the line of *ḥatrum* (mottled zone), descending further to the *Zor* (240 m. b.s.l.). There are many springs in this plain, which ends with a row of truncated hills of Lissan marl. It contains a system of irrigation channels and brooks watering the plain the whole year round. It is densely inhabited, with many sites. The *Ghor* is also a wealthy agricultural region today, where the main crops are irrigated dates and vegetables.

The *Zor* here is narrow and difficult to define, and the eroded hills and ravines of Lissan marl reach almost to the River Jordan. The width of the *Zor* is a result of the meanders of the Jordan: in a place where the river meanders eastward a fairly wide *Zor* is available, and vice versa. The breadth of the *Zor* here varies from 1 km (close to Tell Abu Sus) to several tens of metres (near Maḥdeth Masudiyeh).

Thirty-seven sites were found in both sub-districts in our survey. The



13. Aerial view of Tell el-Ridhghah (Tel Shalem) and the *Ghor*, view to the west, 2013 (A. Solomon). Note the rich agricultural area in the plains.

density of sites in the Bardaleh hills is 2.8 per sq. km, the highest of those described in this volume. The density in the Bardaleh hills, which are much smaller in area, is far greater than in the *Ghor*, which is 1.2 per sq. km. There are two reasons for this phenomenon; firstly, the sites of the Bardaleh hills depended on the agriculture in the *Ghor*; and secondly, the survey in the *Ghor* was not comprehensive. The difficulty of the *Ghor* survey arose because the area was intensively cultivated, probably destroying some of the sites. This is especially so in flat irrigated areas, in which only sites situated on elevations in the plain were found. An example is the site at 'Ein ed-Deir (Site 23), discovered by Mittmann (1970: site 345). When we first reached it in 1991, the remains of the tell were still visible. In 2001, after greenhouses and vegetable gardens were positioned there, there was almost no trace of it. This means that a certain percentage of sites were not found in the cultivated *Ghor*. Support for the assumption that the *Ghor* was cultivated by farmers from other areas can be found in Robinson (1970, III: 31), and also earlier in the Hütteroth and 'Abdulfattah survey (1977: ill. 1), presenting the Ottoman census from the end of the 16th century.

The model of settlement in this landscape unit is notable for the multitude of tells and the stability of the habitation. The tells represent 35% of all sites, an extraordinary phenomenon both in the Manasseh Hill Country region and in the desert fringe lands surrounding the Beit She'an Valley. Of 13 multi-period tells in the area we note Qa'un, Tell el-Ḥammeh, Tell Mukehaz, Tell el-Ridhghah,

Kh. Sakut and the tells over the *Hatrum* (mottled zone) line and the banks of the River Jordan – Tell Jima'in, Tel Tzaf and Tell Abu Sus. One of the most significant is Tell el-Hammeh, identified with Hamat of the Beit She'an stele.

A multitude of tells is characteristic of districts with arable soil and water sources, such as the Beit She'an Valley. The number of archaeological periods in a given site also exceeds the average. In Landscape Unit 27, 201 occupation periods at different sites have been defined, an average of 5.4 occupation periods per site.

The main establishment period here is the Chalcolithic period: eight of the 15 sites that were founded at that period served the bases for most tells in the region. This number, unparalleled in other places we surveyed, testifies to the settlement characteristics of the Ghassulian Chalcolithic culture. Three Chalcolithic sites were found in the complex of Qa'un alone. Sites from that period were also founded in 'Ein Mukehaz and Tell ed-Deir. The 6 ha site in ej-Jobeh (Site 31) belongs to the large settlements of the alluvial fans, together with the sites of Nahal Tirzah and Fazaal (Bar 2014a). The EBA is also present in the Qa'un complex and the majority of the other tells. Small areas of the fortified EBA Ib site near Tell el-Ridhghah (Site 18), located in the alluvial plain north of the tell, have been excavated by Eisenberg (1996). A thick brick wall dated to the EBA Ib was exposed, which means that fortification of sites had already begun then. No rural settlements belonging to the EBA were found, perhaps because they were buried under the alluvium.

The IBA sites are significant. They are not common in the other landscape units in this volume. The eight sites in the Bardaleh Hills are clustered together, similarly to clusters of settlements of the period in the northern Buqei'ah (Zertal 2008: 41-43), and to those near the south of the Beit She'an Valley (Zertal 2008: sites 25, 62, 64 and 66). They all match the settlement profile of the period: large plots of land in the desert fringe areas, and freedom from reliance on flowing water sources. The IBA is not present in the *Ghor* sub-district, possibly because conditions there were less favorable.

Thirteen sites of the MBA II have been found. In almost all tells there is a stratum of the period, yet no single-stratum MBA II sites have been found. The tells have a typical conical shape like those at Hammeh, Abu Sus, Rehov and Beit She'an. Many single-stratum MBA II sites with well developed rural architecture have been found in the desert fringes south of the territory referred to in this volume. The relation of these unwalled sites to the tells is as yet unknown.

The LBA is sparsely represented in this unit, only eight sites being found. Sherds of this period were mainly found in Tell Mukehaz (1)(Site 11) and Tell el-Hammeh (Site 30). It is likely that the Mukehaz site was connected to Hamat (Tell el-Hammeh), an LBA city-state appearing in various sources.

The 15 IA I sites are twice as numerous as in the previous period. Only three

of these were founded in the IA I, and the rest were built in places inhabited centuries before, usually with a settlement gap of various periods. Such a situation, inhabited areas rich with arable land, water and tells, explains the similar number of IA I and IA II settlements in the region (15 sites of the first period compared to 12 of the second). In the other areas of the Central Hill Country, and mainly in the desert fringes, the ratio is approximately 1:2 (for example – 48 IA I sites compared to 86 IA II sites in the eastern valleys and the desert fringes – Zertal 2008: 83-88). The explanation of this situation is that most of the agricultural potential for the establishment of new settlements was already exhausted during IA I, and later settlements continue on the sites of previous ones.

Fifteen Persian period sites were found. There are few sites of this period in eastern Samaria, and almost the entire settlement is concentrated in the Mediterranean districts and the spine areas of the Hill Country (Zertal 1999). Here also no new settlements were founded, and continuation is the rule. The 12 Hellenistic sites suggest a very gradual decline from the previous period, as in most of the Manasseh Hill Country.

There are 11 sites of the ER (Herodian) period, a continuation of the decline mentioned above. However, settlement recovered, and was back to its previous level in the LR (16 sites), and especially the Byzantine (22 sites) periods. In the Byzantine period the settlement reached its peak, corresponding to, and possibly connected to the flourishing city of Scythopolis at the time. Jewish settlement also prospered, as concluded from the Rehov floor mosaic inscription and from other data (Sussmann 1981).

A decrease is evident in the EM period (15 sites), possibly connected to the decline of Scythopolis after the earthquake of 749 CE (Mazar et al. 1993: 223). During medieval times there is no clear change (13 sites). A sharp settlement decline began in the Early Ottoman period (5 sites), expressed in the Ottoman census of 1596 (Hütteroth and 'Abdulfattah 1977), from which one learns that the region was almost deserted, and dominated by nomad tribes.

Summing up, the south of Beit She'an Valley is more an extension of the Beit She'an Valley than an integral part of the desert fringes of Samaria or the narrow Jordan Valley.

Landscape Unit 28 – The Narrow Jordan Valley

The narrow Jordan Valley is a long strip west of the River Jordan that varies in width from 500 m (in the vicinity of Kh. Juneidiyeh) to 2.5 km (east of el-Makhruq). Its eastern boundary is the River Jordan. In the west the boundary runs along the Beit She'an–Jericho road. Occasionally the boundary deviates a little westward, mainly in the places in which non-surveyed enclaves were left



14. River Jordan and the *Zor* in the area of Tell Abu Sidreh, view to the north, 2002 (M. Einav).

when working on Volume 2 of the Survey.

The space between the foothills of Eastern Samaria and the Jordan River is very narrow in most places. This space was created by the westward movement of the River Jordan caused by the rivers and streams descending from east of the Jordan. At the south of this strip, close to Ras el-Kharubeh, the valley widens to become the Argaman plain.

Landscape Unit 28 includes two strips: the *Ghor* and the *Zor*. They are the result of the width of the Jordan channel varying over geological times. The width of the *Ghor* here, made mostly of Lissan marl, averages 1 km. It is an almost level plain cut by deep ravines. Most of these descend from the desert fringe plateau.

This eroded and cut landscape is the result of the considerable variations in elevation and the nature of the ground. The *Zor* is a flat plain, descending gently eastward to the River Jordan, whose bed is at the centre of the *Zor*. The *Zor* is lower than the *Ghor* by 10 m on average. It is made of alluvium and typified by dense vegetation on the banks. Parts of it are currently under cultivation, and others are waste, with annual vegetation growing in them. The access down from the *Ghor* to the *Zor* is by cuttings in the steep slope of the *Zor* hills.

South of the village of en-Na'ageh is a marshy plain spreading from the foot of the eastern Samaria hills (Naqeb el-Baqqar and Ras el-Kharubeh ranges) and the *Ghor*. The especially gentle slope here has poor drainage, and therefore substantial parts of the plain are flooded during the winter. Vegetables and

date-palms are now grown in the district.

The Lissan marl soil in this strip of land of the Jordan Valley is chalky and poor in organic matter, badly cut up, and there are few water sources. Along it are three springs: 'Ein Juneidiyeh, 'Ein Saleh and 'Ein Abu Sidreh, which produce limited amounts of water. Moderately-sized settlements developed next to the springs.

The Israeli security fence, generally adjacent to the Valley road, and the minefields near the border, hampered the continuation of the archaeological survey to the east. Access to this area nowadays is almost impossible. Only in the winter of 2002 were we able to travel along the *Zor*, when we sighted Tell Abu Sidreh, visited Kh. Juneidiyeh, and discovered Kh. Sadd el-Balqawi (Site 41). It is probable that there are additional sites in these problematic areas.

The Beit She'an-Jericho road did not run through the *Zor*, which used to be subjected to frequent floods, but over the shoulder of the *Ghor*, although there also the many wadis transecting the *Ghor* made the passage difficult. The Roman road in Peutinger's Map, few remains of which have been preserved, ran along the very route of the modern road.

Landscape Unit 28 sites have been discovered on both sides of the Valley road. The small valleys to the east and west of the road have also been checked, and the area was surveyed wherever there was no risk of mines.

The 40 sites in this landscape unit slightly exceed the number of those in Unit 27, but the differences between the units are considerable. The area of Landscape Unit 28 is about 50 sq. km, and the density of sites is 1.25 per sq. km. The limited area of the sites here should be noted: most of them are very small, and possibly were seasonal. In comparison to the main site types some general observations are worth noting:

1. Tells. Unlike the Beit She'an Valley, there are no significant tells in the area due to harsh living conditions. The sole tell, Abu Sidreh (Site 45) is small in area and rises at the south of the *Zor*.
2. Cave sites. 13 sites were found, typical of the Jordan Valley. A cave site is a combination of human habitation together with holding animals in a natural cave. Normally, there is a constructed yard in front of the cave, used as an enclosure for the animals and a place for cooking and food preparation for the family. The cave sites were created and put into use due to the special conditions in the Valley. The mountains of the desert fringes descend to the *Ghor* in a series of short cliffs with many caves. Most of them are made of soft conglomerates, not of hard limestone; causing collapses of their ceilings and shrinkage of the cave areas, resulting in caves of modest size with exposed walls. Most of the caves and other rocky shelters along the Valley were settled at various times. Human habitation in these caves was common, beginning in prehistoric times,

and has continued until now. We made an effort to document some of the caves, but could not enter them all.

3. Enclosures. Eight sites, all founded in the IA I or II periods. This is only a small fraction of the main concentration of enclosures around and south of the Sartaba range. The enclosure is defined as an area encircled by a wall, mainly meant for the maintenance of domesticated animals, and rarely used for assembly. The types of enclosures are many and varied. In most cases the inner enclosure area is empty; however, large enclosures occasionally contain inner walls and constructions. The enclosure sites are divided into two groups: round ones, probably used as a base for flock pens, and other more developed ones. To the first group belong, for example, Siret el-Ma'azeb (1) (Site 67), Wadi el-Mughareb (Site 69), Tellul el-Bedha (1) (Site 71), and others. The enclosures at Bedhat esh-Sha'ab (Site 65) and Yafit (3) (Site 105) belong to the second group. These sites belong to the 'foot-shaped enclosure' sites, in our opinion intended for a special function (see Appendixes B and C). The unusual outer shape of these enclosures and their inner arrangement invests them with a certain role in the IA I, and perhaps also in IA II society.
4. Courtyards. Six sites belong to this group, more common in the desert fringes, mainly in the later periods (LR and Byzantine – for comparisons see Eitam 1996: 698-699). The rectangular courtyards designated for the upkeep of animals, mainly cattle, and agricultural products, succeeded the circular enclosures of the IA.

In the 40 sites of this unit, 153 occupation periods at different sites have been defined, or on average, 3.8 occupation periods per site, a much smaller number than that of the other units in this volume. Of this total, periods of peak population are the IA I (15 sites), IA II (29 sites) and LR-Byzantine (22 and 25 sites respectively). Such a distribution accords with the settlement in marginal zones and the desert fringes of Samaria (see Zertal 2008: 83-88). These waves of settlement are characteristic of the Manasseh Hill Country (and presumably also the other parts of the Central Hill Country, the knowledge of which is incomplete). The Chalcolithic period and the Bronze Age are sparsely represented (four sites each), and there are only two sites from the IBA and the MBA, and four LBA sites. A notable dearth is also evident in the Persian and Hellenistic periods (one and three sites respectively). The ER shows a slight increase (seven sites) before the peak of the LR and Byzantine periods. A decline occurred in the EM period (seven sites) before a recovery in the MA (17 sites) and the final decline of the Ottoman period (only two sites).

Landscape Unit 29 – Nahal Tirzah Floodplain

This unit enjoys unique characteristics. Its boundaries are determined by the wadi floodplain, starting from where it widens on its way eastward after the Bab ed-Dayyq bottle-neck.

The unit contains two sub-units, totalling 50 sq. km. The floodplain itself is a long narrow triangle about 10 km long and 2 km wide. South of this triangle, east of the Beit She'an–Jericho road, is situated a large area of the *Ghor* and the *Zor* with an area of about 30 sq. km.

Following are the details of the two sub-units:

The upper north-western part of the floodplain is bounded by the Shechem–Jiftlik road. The area of the triangle between Ras el-Kharubeh in the north-east and the ridges of 'Iraq er-Resifeh in the south-west, is about 13 sq. km. It is a flat floodplain, through the centre of which runs the bed of Nahal Tirzah. The whole area is currently being cultivated, and it is probable that this was so in the past. There are irrigated vegetable fields worked by tenants who inhabit the strip along the Ras el-Kharubeh ridge and below it along the road and in the centre of the area. There are many aqueducts for irrigation purposes; these were probably the drinking water sources for humans and herds in ancient times.

It seems that there were settlements in the alluvial plain itself during most periods, as is proven by our survey among the vegetable fields. Sherds of various periods were found, but the cultivation was too dense to allow definition of settlements or archaeological periods. In this plain is the large settlement of Qarawet et-Tahta (Roman-Byzantine Coraea – Site 85), of which Tell eṣ-Ṣimadi (Site 84) is a part.

The ridge of Ras el-Kharubeh slopes steeply down to Nahal Tirzah and to the edges of the valley, which are very narrow between the road and the foot of the ridge. A large settlement developed in the Chalcolithic period on these edges (now settled by the Jiftlik tenants), and probably spread over the entire area of the floodplain to the riverbed: the limits of this area are not known.

The opposite margins at the south-eastern part of the triangle (Massua district), 4 to 5 km long, are wider and convenient for settlement. It is a relatively arid area with stony-desert ground. The ridges here are lower and indented without a sheer step like that of Ras el-Kharubeh. The alluvial fans of Wadis Zeith, Jaber, Umm Hajer and Jeruzaliyeh exit into the valley. In each one of them are dozens of small ravines merging into the Nahal Tirzah plain. Spacious IBA sites and varied IA enclosures developed along the southern and eastern edges.

The second sub-unit is the alluvium plains of the Jordan Valley and Nahal Tirzah, an area of approximately 30 sq. km. For this volume we have annexed



15. The Nahal Tirzah floodplain, view to the east, 1992. The Sartaba range is in the background (M. Einav).

the area to the Nahal Tirzah floodplain. The Nahal Tirzah riverbed deviates southward and crosses the plain east of the Sartaba from north to south on its approach into the Jordan. The riverbed in this section is 7 km long, with meanders, and flows generally parallel to the Jordan until it empties into it at coordinates 162●/199●. Between the two rivers, the triangular *Ghor* gap formed as an independent landscape unit. This flat district contained four longitudinal strips of land oriented from north to south (similar to those in the vicinity of Argaman in Landscape Unit 28). In the westerly one are included the eastern edges of the Sartaba, made of broad alluvium fans. Among them are the unnamed wadis descending from the Sartaba and Jebel Mastara, from Wadi Jeruzaliyeh in the north to the 'Valley Monument' in the south. These borders are also covered by stony-desert soils. The second strip is a marshy plain of marl alluvium soil 2 km wide. Parts of it are marshlands in which mainly dates and vegetables are grown. The marshy nature of the land made life harsher, resulting in a scarcity of sites. Parts of the area are still covered by marsh plants – common reed and various species of sedges (*Cyperus* spp). Only one site was discovered in the area – el-Marmaleh (Site 103). The third strip is the *Ghor*, 1.5 to 2 km wide. It is very much cut up and noted for its low rounded hills. The soil is marl-bright. Here also the survey was not complete due to the presence of minefields, and no ancient sites have been found. There is modern era agricultural construction: the plain is

crisscrossed by many open concrete water channels. In some places remains of buildings were found with Marseilles tiles dating from 1880 to 1920. These places seem to be farms dating from the end of the 19th to the first half of the 20th centuries. Substantiating the precise dates and nature of these agricultural enterprises requires an additional investigation. The fourth strip is the *Zor*, which we have been unable to survey.

Among the roads greatly influencing the settlement is the main road from Shechem to the Jiftlik (and to Jordan – B₃), and the Beit She'an–Jericho road (B₁₀).

The district is well-watered thanks to Nahal Tirzah. In past times (unlike today) the stream flowed all year round, with the average distance to the sites around it not exceeding 2 km.

The area of Landscape Unit 29 is about 50 sq. km, with 39 sites, a density of 0.8 sites per sq. km. Regarding the types of settlements, the number of tells (two) in this landscape unit is far less than the enclosures (18 out of 39 sites). The two tells are situated in the wadi floodplain, due to the advantageous conditions for their existence there (water supply and arable land). However, the enclosures typify the south-easterly margins from Wadi Hajar in the north to the 'Valley Monument'. Most of the enclosures seem to be seasonal sites connected to grazing. In the Sartaba's eastern margins there is a high density of enclosures, mainly from the IA I.

Also worth noting is the large IBA settlement of Khallet el-Fuleh (Site 81) in the Wadi Zeith floodplain. It spreads across the entire width of the wadi and the hills surrounding it. Its area is estimated to be several dozen hectares, and it is one of the largest of its kind in Israel.

Another type is that of the three forts, established in the IA. Their existence continued during the Roman Period (Rujm es-Si'a – Site 86 and Tell 'Abeid 1 and 2 – Sites 98 and 99). All are located on the lower slopes east of the Sartaba. The first one may have guarded the easterly approach to the Sartaba, and the two others are close to the Beit She'an–Jericho road and might be related to it.

One of the largest sites found in the floodplains is Chalcolithic. It is a combination of three sites which were probably one large alluvial site: Shunet el-Masna'ah, Kaziyet el-Ratrut and Far'at ej-Jiftlik (Sites 77–79). Eight Chalcolithic sites were found, and eight EB I. Then there was a decline to five EB II–III sites; three IBA sites (but one of them is very large – see above); three MBA sites; and not a single LBA presence.

A drastic change occurred in the IA: during IA I, 19 sites were settled in this unit, followed by 16 sites in the IA II. It is interesting to note that this is one of the few regions in the survey where IA I sites outnumber IA II sites, and from this aspect it is also obvious that the strip east of the Sartaba was of special importance during the IA I.

After the IA settlement boost a sharp decline occurred in the following periods (three Persian sites, one Hellenistic site, and three ER sites). Only in the LR period was there a new settlement wave with 21 sites. This wave is known in every unit in the Manasseh area, and the period is the most prosperous time in the history of settlement in the region. A slight decline in the Byzantine period (15 sites) continued in the following periods (seven EM sites, nine MA sites), until declining to almost complete abandonment in the Ottoman period (two sites).

The differences between the sub-districts are also significant. For instance, there are few IA and Roman sites in the Nahal Tirzah triangle, but there is very dense IA and Roman settlement in the eastern margins of the Sartaba.

Landscape Unit 30 – The Sartaba Range and its Surroundings

The Sartaba range is an isolated land block (horst) rising above its surroundings. It is about 15 km long from Bab ed-Dayyq in the north to the 'Valley Monument' in the south. Its maximum width, from Gittit in the west to Masu'a in the east, is 9 km. The area of the entire range, including all relevant districts (the hilly area at its foot in the north, the 'seam' at the west, the hills between Maale Ephraim and Gittit and the Wadi Ahmar valley) is about 130 sq. km, and it is the largest landscape unit included in this volume.

The Sartaba is made of Eocene soft rock lifted by a series of major faults. It contains four rock formations: Zora (chalk and concentrations of flint); Nitzana (chalks with flints and fossilized foraminifera); Horsha (chalk and flint); and Matred (chalks).

The Sartaba massif contains several geological unconformities (Zora formation by the horn of the Sartaba for example). This last indicates a stage of sea retreat and exposure of the ancient rocks to erosion. Another unconformity appears in the Matred formation, probably linked to tectonic activity.

The following stages occur in the region's geological history:

1. Sedimentation of the Judea and Mount Scopus groups, together with the slow rise of the east Samaria anticline during the Senonian and Lower Eocene;
2. Exposure and erosion;
3. Repeated flooding, late Lower Eocene and Middle Eocene sedimentation. The chalks found in the south and north are from this stage;
4. Sea retreat, faulting and fracturing;
5. Flooding and sedimentation of the Argaman member;
6. Fracturing of the Wadi Far'ah system and the sedimentation of terrestrial material – Neogene;
7. The forming of the Jordan Valley rift and Lake Lisan.

We have included several sub-districts surrounding the Sartaba Range in Landscape Unit 30. The similarity of the habitation model in these districts justified their inclusion. These sub-districts are:

1. The Sartaba Heights is the area from Umm Hallal in the north to Tal'at 'Amreh in the south, and from Ras Hāfireh in the west to el-Mastara in the East. It is a relatively flat plateau, of maximum 320 m elevation a.s.l. The original plateau has been fractured and cut by the intensive erosion of Wadis Umm Hājir, Jaber and Jeruzaliyeh (north), Qurein (east), 'Amrah and Ahmar (south), and Khamona and Khellal el-Fal (west). The erosion created high, narrow and long spurs in the Sartaba plateau. These are joined to each other at their bases and separated by deep wadis. These spurs are (from north to south): 'Iraq er-Rasifeh, Lahf el-Laweza, E.P. 244, E.P. 19, el-Mastara, Ras Quneitrah, Ras Hāfireh and Umm Hallal. The extremes in elevation between the wadis and the spurs and the extreme fissuring rendered the heights of the Sartaba almost uninhabitable; and the area was utilized mainly for pasture and the approach to the Sartaba fort.
2. The hills and spurs between the Sartaba and Nahal Tirzah. This large district includes the drainage basins of Wadi Umm Zalin, Wadi ez-Zeith and Wadi edh-Dhba'ah. The area in the vicinities of el-Medakakin, Mughur el-'Avdat and Joret el-Kareairah consists of long shallow spurs separated by small valleys. The traverse from the Sartaba Heights to the northerly zone is an east-west step, called el-Mugheirat in the map. 'Iraq er-Rasifeh is a flat high plateau, isolated from its surroundings by valleys and steep slopes. The high spur of Jebel el-Mahjarah extends north-west from the Sartaba Heights. The hilly district with its small valleys north of the Sartaba is covered by arable soil (terra rossa and Mediterranean forest soil). This was probably the main reason for it being the most populated district in the entire history of the range.
3. The step west of the Sartaba is the juncture between the Sartaba Heights and the alluvial basin of Wadi Afjeam. The step is formed by steep slopes with narrow spurs between them, and with many small wadis descending westward. Along it are three springs ('Ein Abu Daraj, 'Ein Hāfireh and 'Ein el-Mana), but due to the steepness here there are few settlements.
4. The hill district between Maale Ephraim and Gittit extends from the Alon Road in the west to Wadi Afjeam in the east. The district is drained mainly by Wadis Manḥur and es-Sukhun. It consists of long broad spurs covered by good agricultural earth; yet here also there are few settlements.
5. The valley of the Wadi Ahmar area, which drains the south-western part of the Sartaba, is covered by alluvial soil, mainly rendzina. Added to this area is the north bank of Wadi Ahmar from the wadi itself to the heights

of the Sartaba. Despite the small area of the valley (50 ha) it was well settled. The sites in Wadi Ahmar valley have unique characteristics.

The high Sartaba range is encircled by a number of small layer springs: Jeruzaliyeh in the east, 'Ein el-Mana in the south-west and 'Ein es-Sukhun, Hāfireh and Abu Daraj in the west. Their flows are limited, and they were probably used mainly for watering animals. At the top of the Sartaba there is no water, requiring a central water collection system for the fort's needs. Most of the farms and other small sites in the Sartaba range and its environs probably subsisted by collecting water in cisterns.

Roads by-passed the Sartaba range because of its height, steepness and isolation. The Beit She'an–Jericho road (B10) passed to the east, and the two branches of the Roman Fasaelis–Neapolis road (C40) passed to the west. One branch of C40 ascended directly to 'Aqraba, and the other led north along Wadi Afjeam, turning west to Kh. Tana et-Tahta. Here three secondary roads joined it from the east, ascending from the east from the Jordan crossing in the vicinity of the Jiftlik: one via 'Iraq Rasifeh, the second via Wadi Zeith, and the third via Wadi Jaber.

The Sartaba Plateau was reached by a number of roads. The path through Wadi Jeruzaliyeh, which probably served as one of the main approaches to the site; a path along the crest to the east originating in the saddle of el-Mastara; a road, part of which was found by Meshel (1970: 105), that possibly ascended from the south-east through Wadi Qurein; and a road south of the fortress, over the plateau, where cut stairs have been found, probably part of yet another road ascending from the south.

With regard to the types of settlements, 108 sites have been found in the Sartaba range, an average density of one settlement per 1.2 sq. km. This is a low average, but this impression is slightly misleading, since the Sartaba Heights area itself has almost no settlements except for the Alexandrion fortress (and a few sites surrounding it), while the other sub-districts have many sites. The main characteristics of the habitation model in the Sartaba range are:

- All sites are small (500 to 2000 sq. m), and most are one- or two-period farms, enclosures or camping sites;
- There are no tells, and very few multi-period sites;
- Settlement peaked in two main periods – IA II (60 sites) and LR (81 sites). Such a concentration of habitation requires a historical or socio-economic explanation;
- Bronze Age sites are almost completely absent;
- Many of the sites are seasonal;
- The Ahmar valley offers a unique settlement model;



16. Aerial view of the Sartaba fortress and range. Note the hilly terrain, view to the west, 2014 (A. Solomon).

All the sites had the same basic characteristics. In most of the settlements there was a dwelling with an enclosure or courtyard for animals. The differences between the sites, ranging from small farms, such as el-Medakakin (Site 117), to a more developed site with many installations, such as Joret el-Kareairah (Site 115), depended on the development of the individual settlement.

The relatively large courtyards and enclosures suggest cattle and/or flock husbandry. In the small valleys cereals were grown in good seasons, as evidenced by the many threshing floors and installations for the processing of cereals in or near many of the sites; and it is also possible that olive-growing existed here (judging from the installations found, see Eitam 1996: 696-698).

In some instances only a courtyard or an enclosure was found, with no residential structures. This indicates that animals were kept in separate concentrations without direct contact with human living quarters, the use of perishable materials, such as tents, in the construction of dwellings, or the use of these structures to handle agricultural products. The abundance of sites of this type leads to the assumption that this region was a preferred grazing area.

The dating of the majority of sites to two periods, IA II and LR, is typical of the desert fringes of Manasseh. An identical phenomenon has also been found in the eastern valleys and the desert fringes of Samaria, included in Volume 2 of the Survey (Zertal 2008: 85-88, 95-96). In this respect the Sartaba range is similar to most of the 'Samaritan Desert'. For the major waves of settlement in the district certain political circumstances were required to promote settlement.

The subject has already been discussed (Zertal 2001), and will be discussed further in the next section.

In comparison to the IA, the lack of settlement during the Bronze Ages is significant. Many MBA II sites are situated several kilometres west of the Sartaba, in the vicinity of 'Aqraba-Gittit, and it is possible that the site at Jebel el-Mahjarah (2)(Site 130) is connected to them. From the mode of settlement it seems that the considerations of the MBA and LBA settlers were mainly connected to Mediterranean agriculture.

The large number of forts indicates the military nature of the area of the Sartaba, connected to the fortress of Alexandrion. In two cases – Ruġm es-Si'a (Site 86) and 'Ein Abu Daraj Fort (Site 154) – Roman forts were found on top of IA II sites, whose characteristics are unknown.

Settlement classification according to sub-districts is as follows:

- The upper level of the Sartaba is empty, except for the fortress itself, Lower Sartaba (Site 172), Kh. Koufa (Site 173[1]) and a concentration of tumuli at Ras Quneitra (Site 164).
- More than one-third of the sites, mostly IA II and LR farms and enclosures, are concentrated in the northern hill district. The farms have a definite pattern: a dwelling structure and a courtyard used for animals and processing and storage of produce. In certain places (Joret el-Karai-reh – Site 115) systems of quarried installations, such as winepresses and basins for grinding, have been found. Rectangular courtyards from the LR and Byzantine periods, designed as facilities for cattle husbandry (Eitam 1996: 698-699) or agricultural product storage, are also numerous. A concentration of farms and enclosures has also been found on the sheltered 'Iraq er-Resifeh plateau. The stone dam constructed in the Meḥora Valley, close to Mughur el-'Avdat (Site 116[1]), seems to have been part of an irrigation system supplying water to the valley.
- Several structures resembling forts guarding the road have been found along the Roman Jiftlik-Ṭana et-Tahta road (C35). In most of them there is a single fortified structure with a small courtyard, not identified as an ordinary farm. The similarity between them and their distribution along the route of the road, support the interpretation as military structures.
- The western 'seam' between the Sartaba and Wadi Afjeam. Here also, farms and courtyards have been found on the steep western slope of the Sartaba plateau. The springs of 'Ein Abu Daraj and 'Ein Ḥafireh are gathering places for watering flocks until this very day. Unique to the area are the dry masonry structures spread over the spurs close to and around 'Ein Abu Daraj. We found ten of them, consisting of several rooms well constructed with medium-sized stones. The openings are also finely built. The average size of the buildings is about 10 m × 4-5 m. The buildings

were found without roofs, and no sherds or any other means for dating have been traced. They could be modern, but we have failed to establish when and by whom they were built.

- The hill district between Maale Ephraim and Gittit is an almost deserted area with few sites, mainly courtyards and farms. Three sites in the Tawil el-Minyah area at the west of the district were built during the MBA II (Sites 159-161). They seem to be easterly spurs of the groups in the Gittit-Aqraba-Beit Dajan areas. These unfortified settlements are built of large stones, in a masonry style unknown in other places. There is a Neolithic site at 'Ein es-Sukhun (Site 175 – Nadel et al. 2000), next to the spring. Not far from it is the courtyard site of Mrah es-Sahleh (Site 167), built over IA remains.
- Wadi Ahmar valley. There is a considerable concentration of sites in this small district. The reason is probably connected to the streams discharging into Wadi Ahmar – Wadi Sadi and Wadi Zaqaska – which supplied water and alluvium. Another probable factor could have been the shelter from the winter winds provided by the cliffs in the west and the Sartaba plateau in the east. Perhaps the accessibility to the vast pasture lands in the Jordan Valley encouraged settlement. Most of the settlements are divided into two groups: enclosure sites and built sites with no sherds. Many enclosures of different sizes are concentrated in a relatively small area. Most are circular, 20 to 30 m diameter. They are encircled by a single or double row stone wall with additional small structures spread irregularly around. The small number of sherds suggests mainly animal husbandry, and much less use as living quarters. We found a few settlements of the other type, the built-up structure. All contain stone structures arranged in rows or a 'chain' arrangement. The houses, of the broad house type with several rooms, are located at the foot of a slope or in the plain. The architecture is fairly advanced, but there are no flint or basalt finds, and almost no sherds. The few that were found are dated to the Chalcolithic, IBA and MBA II. This concentration of sites in Wadi Ahmar and their special properties require additional research.

In the 108 sites of this unit, 369 occupation periods at different sites have been defined as detailed in Table 3. The main points of interest are:

- 29 prehistoric sites (in the other three units there are only eight prehistoric sites altogether). Especially interesting is the habitation in the Middle Palaeolithic period (see Appendix A for more details).
- The absence of Bronze Age finds.
- The IA I saw a gradual change, with 20 sites; while a settlement boom occurred in the following IA II with 60 sites.

- After the IA II habitation an almost complete neglect of the area in the Persian period followed (only one site).
- The erection of the Sartaba fortress aroused new interest in the region, and in the Hellenistic (18 sites) and ER periods (22 sites) the area is again inhabited.

Type of site	Landscape Unit 27 45 sq. km; 37 sites	Landscape Unit 28 50 sq. km; 40 sites	Landscape Unit 29 50 sq. km; 39 sites	Landscape Unit 30 130 sq. km; 108 sites	Total
Tell	13	1	2	0	16
Cemetery	1 (very large-regional)	1 (single tomb)	0	1 (5 tombs)	3
Enclosure/courtyard	1/1	8/6	18/1	30/28	57/36
Dwelling cave	0	13	0	7	20

Table 2: Distribution of types of sites: comparison between the landscape units.

Period	Landscape Unit 27 45 sq. km; 37 sites	Landscape Unit 28 50 sq. km; 40 sites	Landscape Unit 29 50 sq. km; 39 sites	Landscape Unit 30 130 sq. km; 108 sites	Total
Chalc.	15	4	8	11	38
EBA I	10	4	8	4	26
EBA II-III	8	4	5	1	18
IBA	8	2	3	2	15
MBA	13	2	3	6	24
LBA	8	4	0	1	13
IA I	15	15	19	20	69
IA II	12	29	16	60	117
Per	15	1	3	1	20
Hel	12	3	1	18	34
ER	11	7	6	22	46
LR	16	22	21	81	140
Byz	22	25	15	42	104
EM	15	7	7	7	36
MA	13	17	9	45	84
●tt	5	2	2	16	25

Table 3: Distribution of sites according to periods: comparison between the landscape units.

- A new and very significant settlement increase occurred in the LR period. This was the most densely populated period in the region, with 81 different and diverse sites.
- A decline occurred in the Byzantine period (42 sites) due to the abandonment of the Sartaba fortress. The crisis was severe in the EM period (7 sites), and only in the MA did a new relatively small wave of settlement occur (45 sites).
- The area declined again in the Ottoman period (16 sites), a picture well attested in the other units of this volume, and generally in Northern Samaria and the Jordan Valley (see below).

D. THE SETTLEMENT BY PERIODS

The archaeological survey is the basis for the interpretation of settlement layouts and their linkage to history. The information derived from it is primary, does not come from another source, and enables settlement, archaeological and historical analysis. The difficulty in this volume is the division into isolated landscape units, without clear organic connections between them. This division has been taken into account, and the review has been arranged according to the landscape units and the bonds between them.

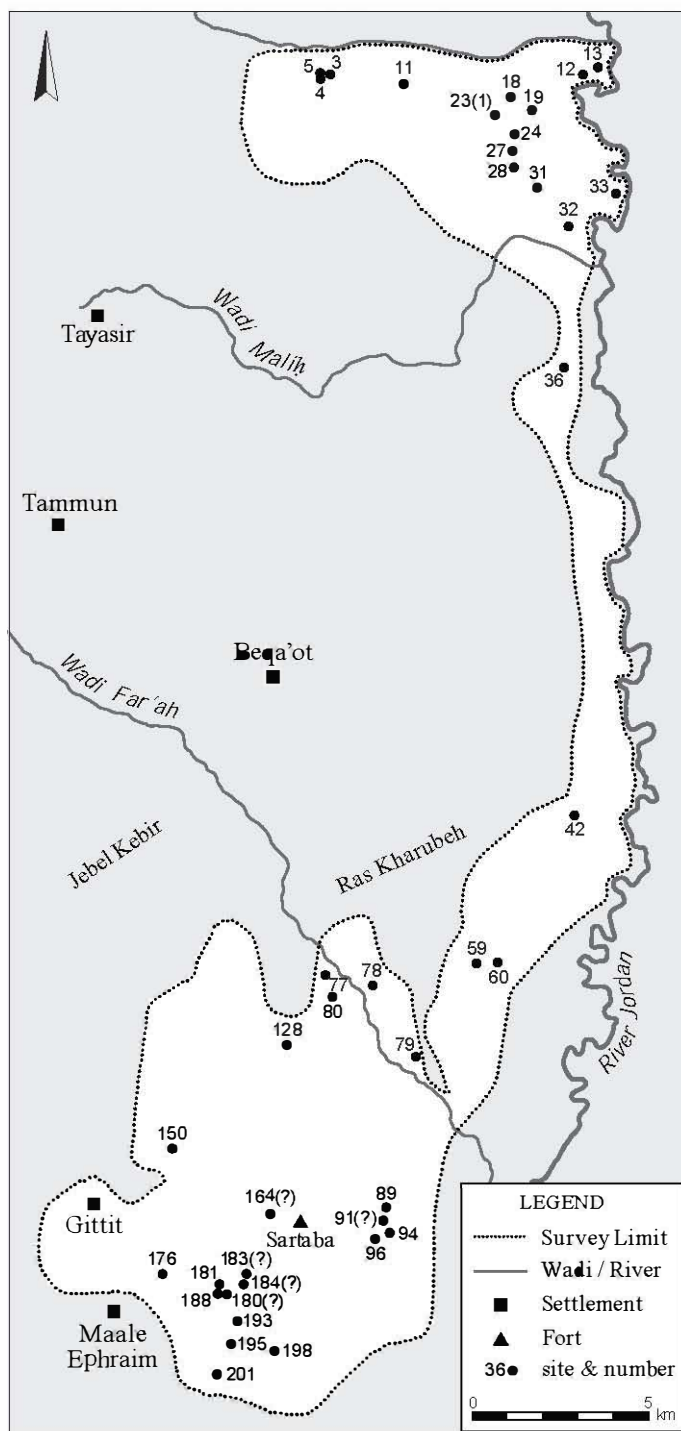
The analysis in this volume is based on 224 sites investigated during the Survey and published here. The prehistoric sites were researched by H. Winter, and are fully presented in Appendix A.

1. The Chalcolithic Period (4500-3800 BCE)

Chalcolithic sherds have been found in 38 sites out of the 224 included here (16.9% of the total). This high percentage reflects the tendency of the inhabitants to settle both in the rich Jordan Valley and in desert fringe areas. These results are supported by additional new studies (Bar 2014a). However, differences between the landscape units are evident: the Chalcolithic sites constitute more than a third (39.4%) of the total number of sites in the Beit She'an Valley, 10% in the narrow Jordan Valley, 20.5% in the Nahal Tirzah floodplain, and 10% in the Sartaba range.

Sherds of the period appear in some of the Chalcolithic tells in the south of the Beit She'an Valley (Qa'un [Site 3], Mukehaz [Site 11], Tzaf [Site 12], Abu Sus [Site 33] and others). This also happened in the Beit She'an Valley proper (Tzori 1962: 197), where the density of Chalcolithic sites is very high.

The main form of settlement in the Chalcolithic sites was the extended site in the floodplains. These were large settlements (up to tens of hectares), spread



17. Map of the Chalcolithic Period sites.

out and unfortified, with their houses scattered across the area. An example is ej-Jobeh (Site 31), where the settlement is spread over a large area in what is now a ploughed cultivated field, while its location had no defensive benefits such as elevation or lookout. Not far from the site is the spring of 'Ein es-Sakut. The pottery and lithic finds in the site are quite abundant (and see additional data in Bar 2014a: site 15).

In the absence of suitable conditions, the Chalcolithic people did not settle in the narrow part of the Jordan Valley. In contrast, there are numerous sites in the Nahal Tirzah floodplain. The large settlements in the northern part of Nahal Tirzah (Shunet el-Masna'ah – Site 77, Kaziyet el-Ratrut – 78, Far'at ej-Jiftlik – 79, etc.) are noteworthy. These form a strip at the foot of the Ras el-Kharubeh ridge, stretching southward, probably as far as the Nahal Tirzah riverbed. It seems that most of the floodplain was settled in this manner.

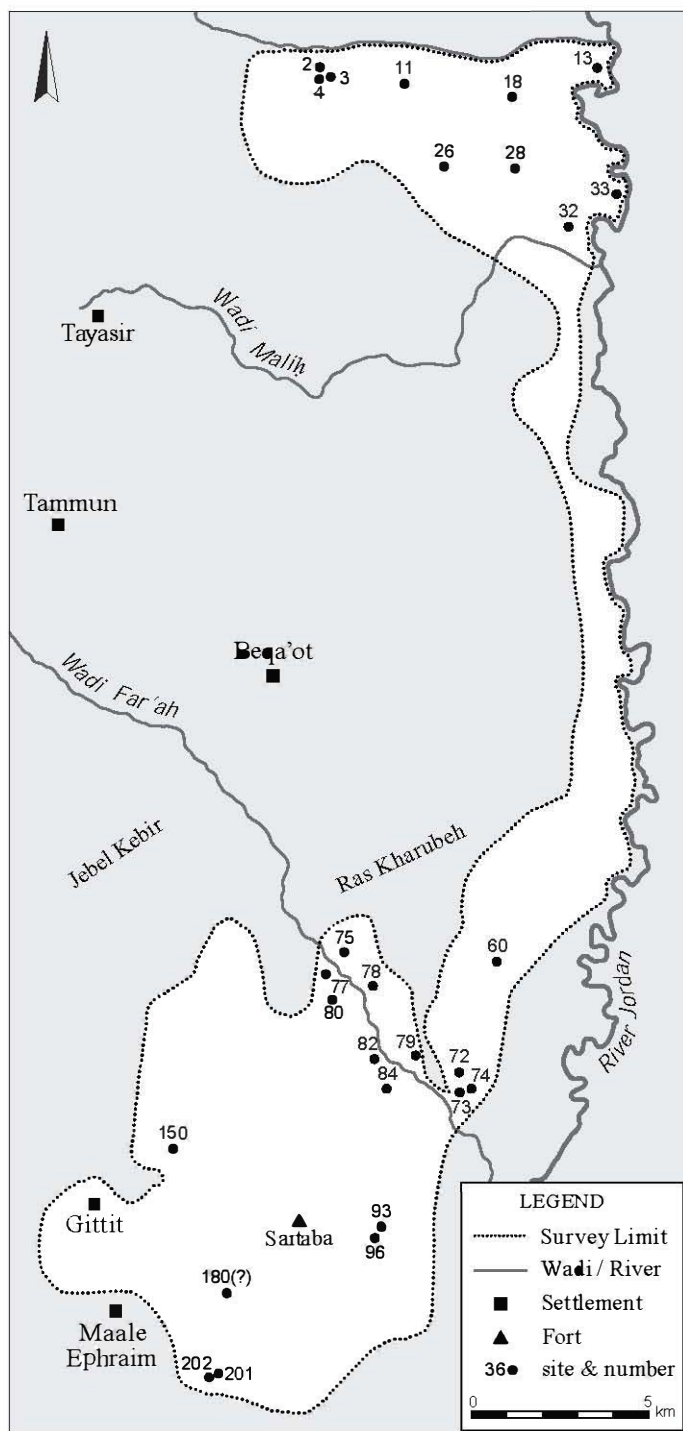
Another type is the grazing enclosure, on the rocky spurs descending into the valley and in the wadi openings. These are simple stone circles, occasionally with a few structures attached to them. They were probably used as seasonal settlements belonging to the alluvium plain settlements, part of whose economy was based on grazing. Such are, for example, the Massu'a sites (Sites 89, 91, 94-96). The problem with these sites is the meagre amount of identified pottery: therefore, their dating to the Chalcolithic period should be regarded as preliminary (Bar 2014a: 62).

In comparison to the eastern valleys and the desert fringes (Zertal 2008: 69-71), grazing enclosures are also mentioned in Volume 2 (Zertal 2008: 71), but evidence of the large alluvial plain sites is lacking.

For a detailed study of the Chalcolithic period in this region see Bar 2014a.

2. The Early Bronze Age Period (3800-2500 BCE)

The EBA is notable for a distinct change from the EB I to the EB II-III periods. While the EB I seems to be a continuation of the Chalcolithic settlement pattern (especially the settlements in the large valleys of Beit She'an and Tirzah), the EB II-III is noted for a gradual decline in settlement activity. In this volume we recorded 26 EB I and 18 EB II-III sites, together comprising 19.6% of the total. There is an interesting contradiction between the dearth of sites and the EB II series of fortified tells found in Wadi Far'ah and along the line of Tell Za'anuni (Bar 2014a: site 42) – Kh. Yannun – Tell Juraish (Bar 2014a: site 66), and from there further south. All these tells are located at the boundary of the desert fringes, about 10 km west of the River Jordan, and all are dependent on springs for water. There are almost no EBA sites east of this line of tells, although there are large sites of the period in Jericho and Beit She'an. Another series of tells and sites is found in Wadi Far'ah itself (Zertal 2008: 71-75). However, the Jordan



18. Map of the Early Bronze Age Period sites.

Valley proper is almost devoid of settlement at this time, despite the availability of water sources. There is a sort of 'geographical-settlement gap'. The reason for this is unclear, and it seems that the environmental or political conditions did not encourage EBA II-III settlement there (see Bar 2013a for additional data).

For a detailed study on the EBA I period in this region see Bar 2014a.

3. The Intermediate Bronze Age Period (Middle Bronze I; Early Bronze Age IV; 2500-2000 BCE)

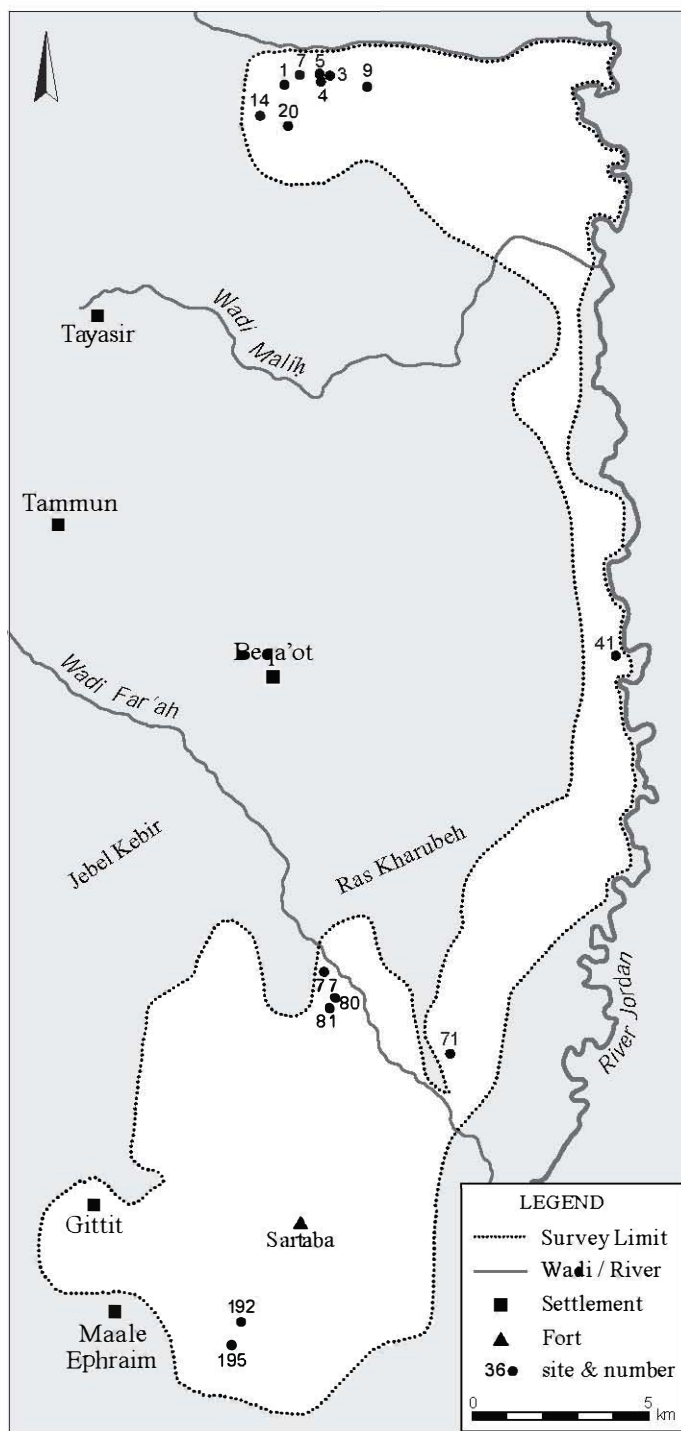
This period also is noted for its sparse habitation, in which eastern Samaria is almost completely empty (for a general review see Palumbo 1990). Only 15 sites (6.7% of the total) with pottery of the period have been found, eight of them in the south of the Beit She'an Valley.

The largest group in Landscape Unit 27 is clustered around the Qa'un springs and Bardaleh plain. The representative site of the IBA is situated in the plain: it is unfortified and its structures are dispersed over the area. The plans of the houses are difficult to distinguish because agricultural activities have destroyed most of them. On the other hand, the abundance of pottery and lithic finds leaves no doubt as to the existence and size of the settlements. Additional evidence of the wealth of the population is the large cemetery in Qa'un (Site 4), in which the shaft-tombs are of a type well known in cemeteries all around the country.

The cemetery at Qa'un has been dug up by grave robbers. This sad fact enabled us to investigate the structure and form of burial. The tombs are spread along a low rocky cliff at the top of a slope, overlooking Tell Qa'un from the east. The entire soft limestone slope, at least 1000 m long, was dug out for burial purposes. The tombs are arranged 2–3 m apart in irregular rows over the slope. The entry to the tombs is by shafts averaging 1 m in diameter and 1–2 m deep. The burial chambers are square or rectangular. Tombs also have rolling stones found blocking their entrances. Outside the tombs were found large pieces of pottery from several periods, testifying that this cemetery served over a long period.

There are only two sites from the period between the southern Beit She'an Valley and Nahal Tirzah. For example at Khirbet Wadi Sadd el-Balqawi (Site 41), on the shoulder of the *Ghor*, we found a small site with an abundance of pottery. No architecture was traced, but the exploration was not completed due to the danger of mines.

In contrast, Khallet el-Fuleh (Site 81), near Nahal Tirzah, is one of the largest IBA sites in the country. It is located at the exit of Wadi Zeith into the Nahal Tirzah floodplain. In the estuary of the dry wadi, which is at least 1 km wide, remains of structures with floors, walls and an abundance of sherds were



19. Map of the Intermediate Bronze Age Period sites.

found. The unfortified and dispersed settlement was built on low terraces on the moderate slope of the estuary. Most of the houses are well built single room broad-houses with supporting walls.

A concentration of IBA sites was discovered by Glueck (1951: 322-329) in the Sakut Valley east of the Jordan. There were several large sites, particularly the group of Umm Hamad el-Gharbi. According to Glueck, the large site in the east, opposite the confluence of Nahal Tirzah with the Jordan, was established on virgin soil. This fact characterizes most of the IBA sites on the western side of the Jordan. There is a resemblance between the sites in Wadi Zeith and Umm Hamad el-Gharbi, and it is possible that the settlements on both sides of the Jordan were interconnected economically and socially.

We have not yet found many IBA sites south of Nahal Tirzah. In Jericho there were large cemeteries around the tell, with only a small settlement on it (Kenyon 1993: 679). With the information in Volumes 2 and 4, and the new research initiated by Bar at the fortified site of Kh. el-Meiyiteh (Bar et al. 2013; Bar 2015a; 2015b), and the rural site of E.P. -167 (Bar 2012; Appendix D), new data and understandings of this period are coming to light. Worth noting is the similarity between the two banks of the River Jordan and the existence of fortified sites west of it.

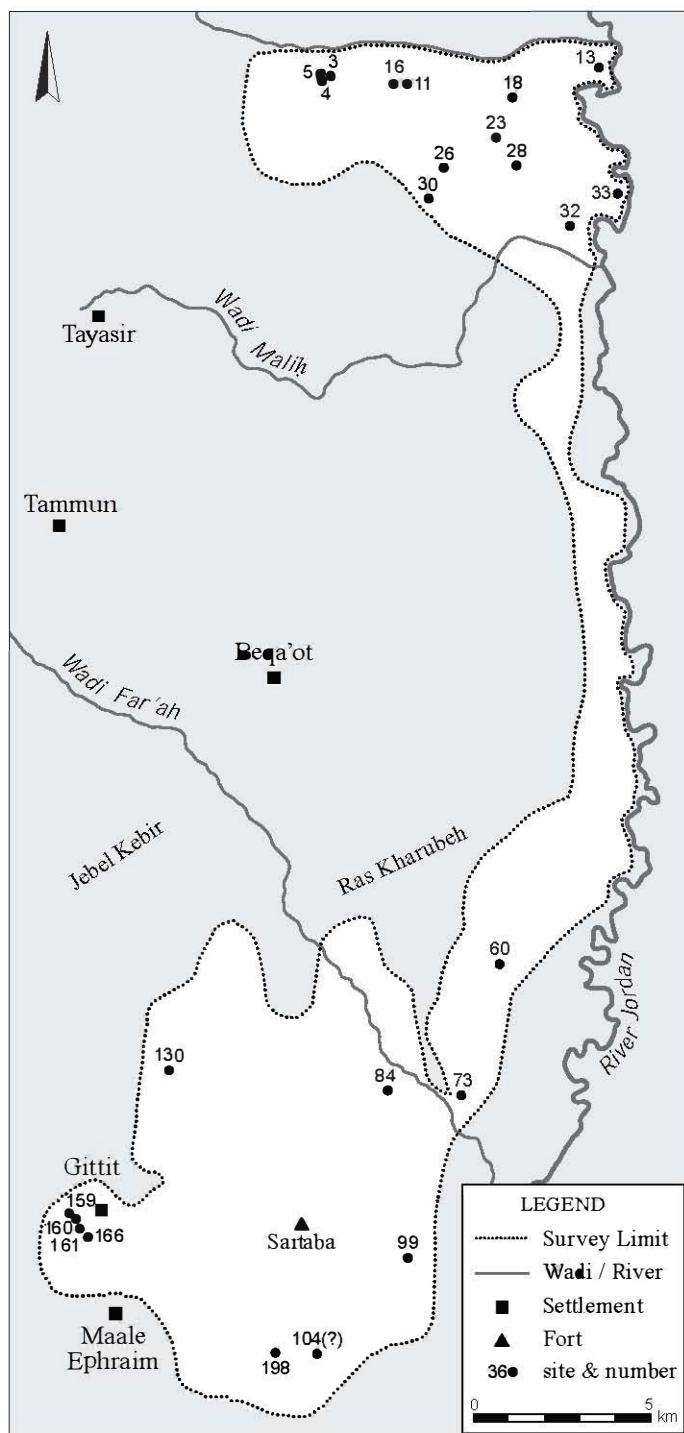
4. The Middle Bronze Age II Period (2000-1550 BCE)

The 24 sites of this period, comprising 10.7% of the total, are few in number in comparison to the other parts of the Manasseh Hill Country. For example, in the eastern valleys which are next to the areas included in this volume, 63 MBA sites have been found, amounting to 23% of the region's total number of sites (Zertal 2008: 78). It seems as if the settlement wave of the MBA IIb had stopped short in the eastern valleys and in the areas west of the Jordan Valley, resulting in a modest dispersion of sites.

In the south of Beit She'an Valley 13 sites have been found: almost all, including Qa'un (Site 3), Mukehaz (Site 11), Jima'in (Site 13), Hammeh (Site 30) and Abu Sus (Site 33), are small or medium-sized tells. Some were founded during MBA II, and others already during the EBA or Chalcolithic periods.

Apart from Kh. es-Sakut (Site 32) and Abu Sus (Site 33), the tells remained small, and did not develop into real cities. Possibly this is connected to the two neighbouring important sites of Rehov and Beit She'an. Seemingly the existence of these sites hindered the development and growth of other settlements. This is supported by the fact that most tells in the Beit She'an Valley are generally small – 3000 to 5000 sq. m, and rise only slightly above the plain.

Only two sites from this period were found in the narrow Jordan Valley: Zbeidat (Site 60) and el-Makhruq (2) (Site 73). In the Nahal Tirzah floodplain



20. Map of the Middle Bronze Age II Period sites.

MBA IIB sherds were found only in Tell eṣ-Ṣimadi (Site 84). A few sherds were also found in Tell 'Abeid (2) (Site 99) and Yafit (4) (Site 104).

The six sites in the Sartaba range are an echo of the presence west of it in the areas of Gittit, Kh. Ṭana et-Tahta, and Beit Dajan. The temenos site, with its possible cultic nature, in Jebel el-Mahjarah (2) (Site 130), was founded during the MBA, and continued in use afterwards. The trio of sites at Kh. Tawil (1) – (3) (Sites 159–161) and that of Wadi el-Manḥur (1) (Site 166) are offshoots of the large MBA entity in the area of Gittit-Yannun.

5. The Late Bronze Age Period (1550-1200 BCE)

Eight of the 13 sites of the period are located in the south of the Beit She'an Valley. Sherds were found only in tells such as Qa'un (Site 3), Mukehaz (Site 11), Jima'in (Site 13), Sakut (Site 32) and Abu Sus (Site 33). The situation here is similar to that in the tells of Beit She'an and Ḥammeh (Site 30): where, despite the considerable settlement decline, the existence of the LBA settlements continued.

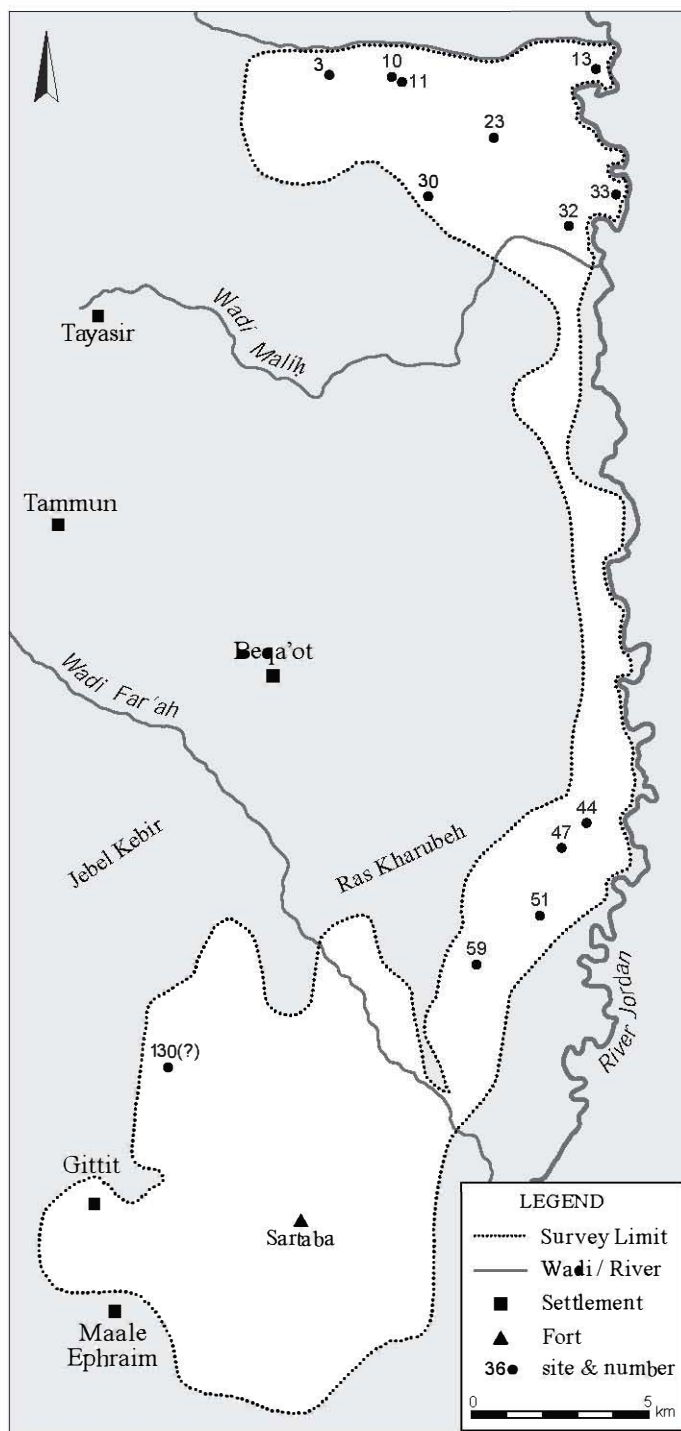
We found sherds in four sites in the narrow Jordan Valley, but none from this period in the Nahal Tirzah floodplain. In the Sartaba range only one site had LBA sherds. All this testifies to an almost complete lack of LBA inhabitants in these areas. In the preceding MBA period the settlement here was relatively sparse, but during the LBA it reached its nadir. The paucity of finds matches the extreme settlement decline during this period, especially in areas far distant from the coastal regions.

6. Iron Age I (1200-1000 BCE)

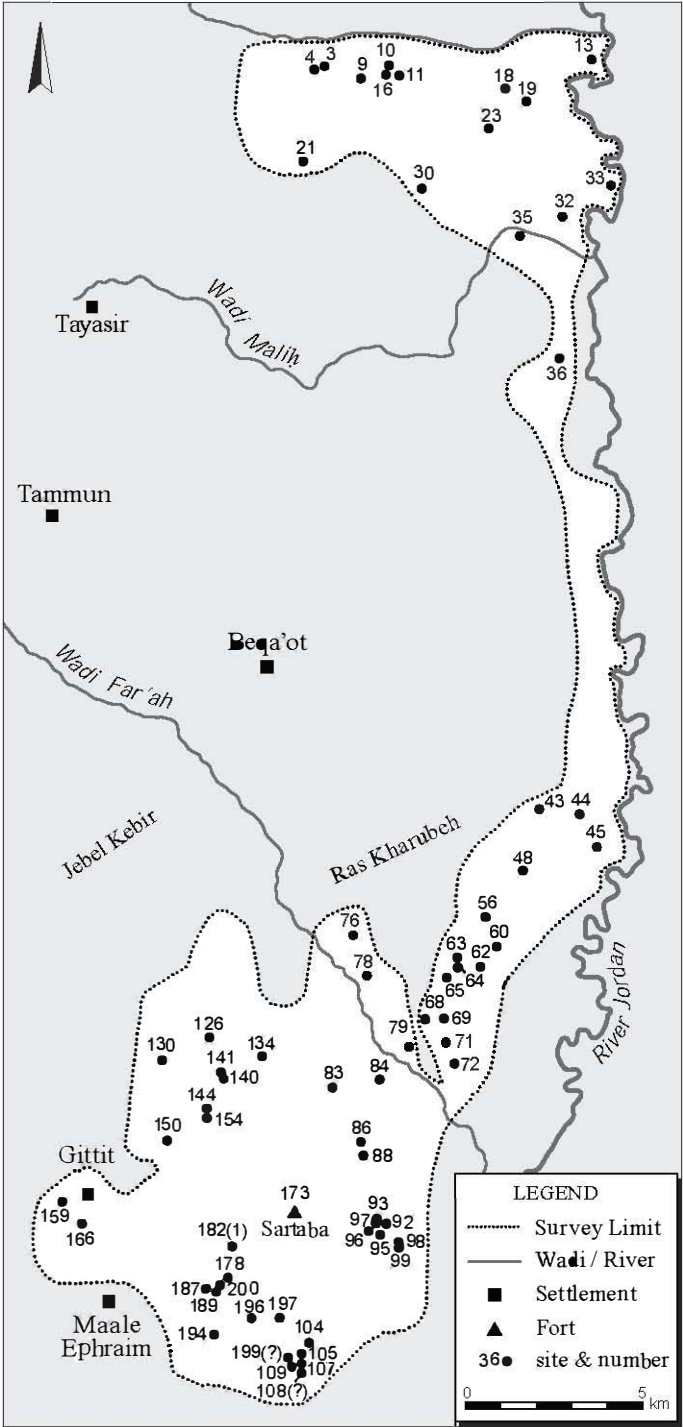
The first large settlement wave started in the area during this period. IAI sherds were found in 69 sites, 30.8% of the total. This is a notable advance from the absence of previous settlement. Almost all sites were founded on virgin soil, without a preceding settlement. This supports, in our opinion, the conception that the settlers originated from outside the region.

Most of the sites are small, and many of them are enclosures. The abundance of enclosures indicates a nomadic or semi-nomadic population, at least at its inception, and it seems that the population composition differed from that of preceding periods. Other common types of sites are small sites or a scatter of sherds without architecture. Such a division points in two directions:

1. In the areas of permanent settlements in this period, there is a certain degree of continuity on the tells; but without excavation it is difficult to determine the character and date of such settlements. In nearby regions tells were excavated in Beit She'an, Ḥammeh (Site 30) and Rehov. In Beit



21. Map of the Late Bronze Age Period sites.



22. Map of the Iron Age I Period sites.

She'an, two IA I layers – Layer VI (13–12th centuries BCE) and V (11–10th centuries BCE – Yadin and Geva 1986), were found and published. Layer VI represents the termination of Egyptian rule (Mazar et al. 1993: 217–218), and is divided into Lower VI and Upper VI. The second stage – Upper VI – is situated on the destruction layer of the Egyptian stronghold, but its nature and characteristics are unclear. The overlying Layer V, which has been related to the 11th century BCE, possibly belongs to the settlement of the Sea Peoples. The material find in this stratum, published by Yadin and Geva (1986), differs considerably from that of the settlement in the Central Hill Country, and the pottery bears typical marks of the LBA culture. Such are, for example, many of the painted and decorated types (Yadin and Geva 1986: Figs. 8:6; 9:9; 27:5, 13 and others). Collar-rim pithoi and other types from the Hill Country repertoire are missing. In Tell el-Hammeh the IA I layers were only superficially excavated (steps J–K, Cahill and Tarler 1993: 562). The pottery we collected dating to the IA I is different from that reported from the tells in the southern Beit She'an Valley, and resembles more that of the Central Hill Country. In Tel Rehov area D at least three IA I strata (Phases D5–D2) were excavated (Mazar 1999). The Pottery is typical IA I with a large amount of painted decorations, mostly red on a buff surface. Red slip and burnish collared rim jars and pithoi are missing. Mazar dated these strata to the period between the end of Egyptian domination in the region (c. 1140 BCE?) and the end of the eleventh century BCE (Ibid: 16).

2. The sites that were founded during this period, in areas void of considerable permanent settlement, might represent a new ethnic or social community, originating from outside the region. It is difficult to identify earlier communities in the surveyed areas which might have been the source for such populations. We cannot see a slow organic evolution here, but a relatively rapid entry of new populations whose origin may be in Jordan. Evidence for this is the similarity between the LBA-IA I pottery types found in the Central East Jordan Valley and that found in the Central Hill Country on the western side (van der Steen 1996: 68).

The IA I enclosures are divided into three groups based on their distribution and types:

1. Circular enclosures. These form more than half of the enclosures, and range in diameter from 25 to 40 m, and it seems that they served mainly as animal pens. The interior is surrounded by a wall up to 1 m wide, made of one or two rows of stones. A few sherds are scattered within or around the enclosures.
2. Composite settlements that include structures with adjacent courtyards.

Six such settlements were found: each one containing up to 20 complexes of structures and courtyards. The structures are small (4×4 m in average) and built of large stones; and the courtyards are rounded and attached to the structures.

3. 'Sandal' or 'foot-shaped' enclosures. Up to now three complete sites have been found, and one or two ruined ones may also belong to this type. These are large sites, ranging from 3000 sq. m to 1.5 ha in area. Their unique shape, resembling a sandal or a foot, is sort of combination of two oval enclosures. In two of the sites the encircling wall was constructed of parallel walls with a gap 1.5–2 m wide between them. This gap between the walls and the paving within it point to a possible walkway, partly or completely encircling the site. In the inner area there are some structures that are not typical dwelling units, alongside paved areas, which prove that the enclosures were not animal pens. In most cases these enclosures were built in the flat plain of the Jordan Valley, in the wadi estuaries near Argaman, and east of the Sartaba. Some are located at the foot of a slope, possibly used for seating an assembly. We can assume that these sites, which were constructed with great care, served for public gatherings or cultic functions. Similar conclusions may be derived from the excavation of the sites at Bedhat esh-Sha'ab and Yafit (3) (see Appendixes B and C).

The distribution of the IA I sites in the area is also significant. Between Argaman and Wadi Ahmar four clusters of geographically separated sites with similar designs were discovered. In each one there are one or two composite settlements, a 'foot-shaped' enclosure and several circular enclosures. The first of these clusters is north of Nahal Tirzah, near Argaman: a large foot-shaped enclosure (Bedhat esh-Sha'ab), with three circular enclosure sites in its vicinity. The three other clusters, also including sites of this type, were found at the foot, east and south of the Sartaba. These are the clusters of Masu'a; Yafit and Wadi Ahmar. In the Masu'a cluster there is a possible foot-shaped enclosure (Masu'a [4] – Site 92), a composite settlement and two circular enclosures. In the Yafit cluster there is a foot-shaped enclosure (Yafit [3] – Site 105), two composite settlements (Yafit [4] and Yafit [5] – Sites 104 and 109) and two circular enclosures. In Wadi Ahmar there are two composite settlements (Wadi Ahmar [4] and Wadi Sadi [2] – Sites 184 and 187) and six circular enclosures.

The material finds in the IA I enclosures and the composite settlements are generally meagre, and in most of them there is no correspondence between the investment in the architecture and the scarcity of pottery. It brings to mind the likelihood of seasonal settlements.

Comparison with other regions shows that the settlement architecture model is different from that accepted up to now. Hertzog (1983), Finkelstein

(1988: 235-250) and others assumed that the initial Israelite settlements included enclosures in which the habitants lived in houses which served as the enclosure wall. This opinion was based upon a small number of sites, such as Izbet Sartah at the boundary of the Plain of Sharon, the Negev sites (which are much later in date), and others. During the Manasseh Hill Country Survey, a different model was found: in the Shechem syncline most of the settlements were small villages with an area of 2000 to 3000 sq. m (Zertal 2004: 55), whose plans differed from the suggested model of the Israelite settlement. An identical situation is found in the area between Nahal 'Iron and Nahal Shechem (Zertal and Mirkam 2016: 40). Even in the eastern valleys and the desert fringes (Zertal 2008: 83) only three or four enclosed settlements of this type have been found (out of 47). Therefore it seems that the IA I settlement structure differs from what is generally considered to be the model of the Israelite settlement. As the Manasseh Hill Country Survey is now the largest area to have been fully surveyed, and most sites in it have been measured, we may suggest two aspects of the initial Israelite settlement architecture. Firstly, most small sites were enclosures and not enclosed settlements; secondly, the division was regional-chronological: enclosures or small rural settlements in the Jordan Valley at the beginning of the Israelite settlement, and clustered villages that were established in the Central Hill Country later in the settlement process. Apparently, most of the enclosures in the Jordan Valley were not dwelling enclosures: most of them were circular pens for holding animals (maybe with tents nearby), and the larger ones were for assembly, cult or other functions unknown at this stage.

7. Iron Age II (Kingdom of Israel, 1000-722 BCE)

The 117 sites from this period demonstrate a further increase in the number of sites and settlement density. This phenomenon is connected to the general policy of the Kingdom of Israel (Samaria): in our opinion, the Kings of Israel devoted great efforts in settling the borderland (Zertal 2001: 44-45). It has been shown that throughout eastern Samaria the site density was greater during the IA II than at any other time except the LR-Byzantine period (see data in Zertal 2008: 85-88).

IA II is notable for the uneven distribution of sites among the various units in the region. Only 12 sites were found in the south of the Beit She'an Valley. This can perhaps be explained by the dense habitation at that period in the other parts of the Beit She'an Valley, and the development of large tells (cities) in that region (Beit She'an, Rehov, Hammeh [Site 30], es-Sakut [Site 32], Abu Sus [Site 33] and others). A significant portion of the population was probably concentrated in these cities, and there were only a few small sites in the open areas. Perhaps the very existence of the large sites did not allow

new rural settlement on a large scale. The Israelite population which did not settle in tells in the easily settled area, established (or continued the existence of) small settlements in unfavorable borderlands. If one divides the narrow Jordan Valley (Landscape Unit 28) into two parts, northern and southern, it will be noted that there was only a small number of settlements in the strip along the Jordan, most of them being small farms. In the south of the unit (the Argaman-Makhruq areas), there is a considerable concentration of sites. This concentration has a special nature – almost all the sites are enclosures, cave settlements or sherd scatters, attesting to nomad encampments (see below).

In the Nahal Tirzah floodplain, 16 sites were found, including Tell eṣ-Ṣimadi (Site 84). The most significant settlement concentration was around the Sartaba range (60 sites), which is divided into three sub-areas: the foot of the Sartaba in the east (the fewest sites); Wadi Ahmar and the hills north of the Sartaba (numerous sites), and the upper part of the Sartaba range, where there are no sites. East of Meḥora there was a great number of small IA sites, mostly farms and enclosures, over which similar sites were established during the LR period. A similar connection between the IA II and the Roman period is also noted in the fringes of Samaria, east of Zababda, Tubas and Shechem (Zertal 2008).

The division according to types of sites during this period also has important implications. We have determined that the fortified tell, the Biblical fortress city, was the central site in a complex of settlements in the Kingdom of Israel. The fortified tells mainly characterize the Mediterranean zone surrounding the city of Samaria. In the desert fringes and in the Jordan Valley the ecological resources were utilized better by the small sites – the farm, enclosure or cave.

Several small fortified tells were found in the south of the Beit She'an Valley, most already founded before the IA (e.g. Qa'un [Site 3], Mukehaz [Site 11], Ridhghah [Site 18] and Abu Sus [Site 33]). Most are small in area, and it is unclear whether they were fortified cities, or possibly (judging by their size) only served as forts. The fortified cities of Beit She'an, and especially Rehov, were the central sites in the valley, and most of the population apparently sheltered in them.

In the *Zor* is the small tell of Abu Sidreh (Site 45), the nature of which we have no idea. The only other tell is Tell eṣ-Ṣimadi (Site 84) in the Nahal Tirzah floodplain. It seems that this site, together with Tell Abu Sus, Sheikh Diab (Zertal 2012a: site 23 and appendix 7), el-Makhruq (Site 73; Zertal 2008: site 269) and others, served mainly as the foundations (podiums) of forts built on them.

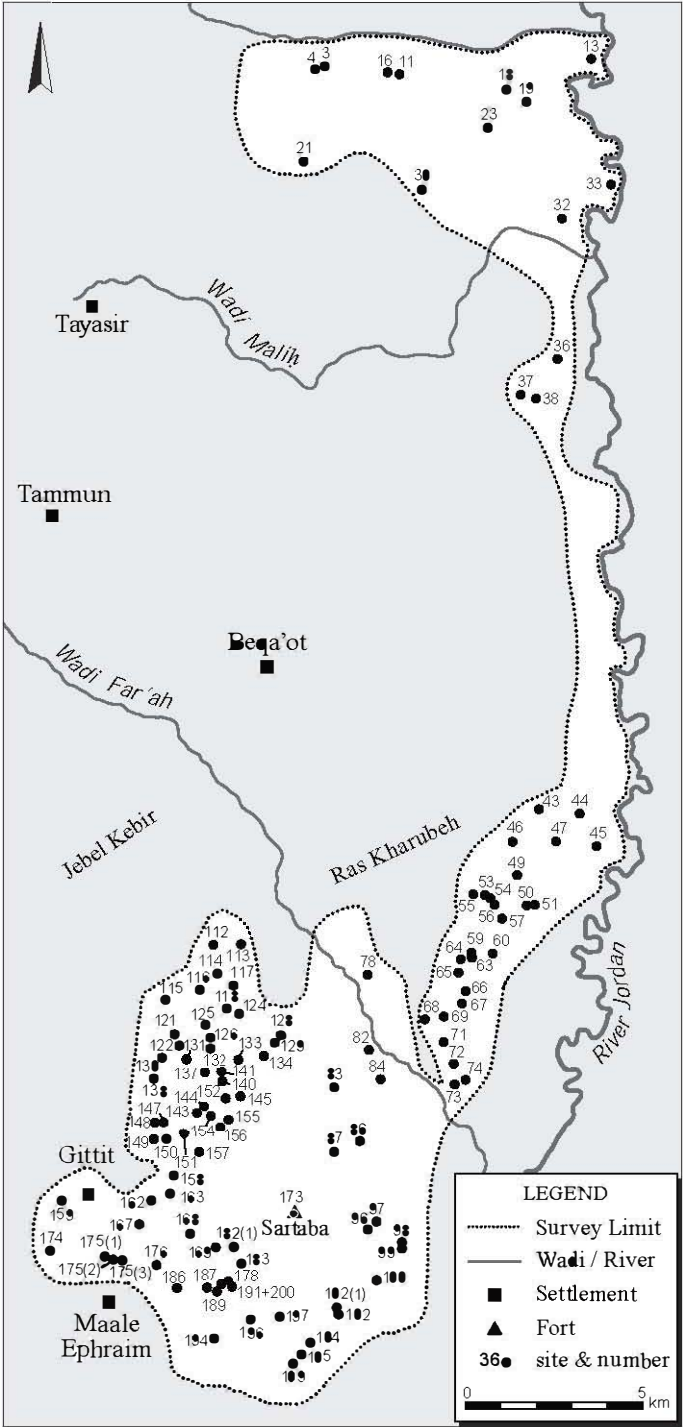
The number of IA II forts is noteworthy. They are divided into three types: oval, circular of the 'Ammonite' type (Rujem Malfuf), and others.

- Two oval type forts have been found – Tell 'Abied (1) and (2) (Sites 98-99)
 - on a shoulder, west and above the Jericho–Beit She'an road (B10). These

are about 50 m long, with an area of 2000–3000 sq. m. They had fairly thick walls, but without excavation we cannot learn more about them. The surroundings of both forts were supported by walls. Also interesting is the existence of these sites, only 300 m apart. It is difficult to decide whether they functioned simultaneously or one after the other. Pairs of forts from this period are typical of the forts of the Negev (Cohen 1993: 1126–1133).

- The **circular** forts of the Rujem Malfuf type have already been published (Zertal 1988; 1995; Yeivin 1992). This is a circular fort (a tower) about 20 m in diameter built from three concentric walls. The spaces between the walls create rooms or cells. This type was first discovered near Amman, Jordan, and published by Mackenzie (1911) and Glueck (1939: 166). The latter suggested the name of 'Ammonite' forts. Such a fort was discovered at el-Makhruq in the Survey area on the ridge between the Jiftlik–Shechem and Jericho–Beit She'an roads (Zertal 2008: site 269). This fort was excavated and published by Yeivin (1992). North of it, at the eastern approach to Wadi Malih, we published an identical fort (Kh. esh-Shaqq – Zertal 2008: site 81), and suggested naming it the 'Tower of the Watchmen' (II Kgs 17, 9; 18, 8). The third fort, Rujm Abu Muhayer, was discovered during the Emergency Survey (Bar-Adon 1972: site 10), and was also excavated by Yeivin (1992: 156–157). The forts mark the origins of the three roads ascending from Jordan and the Jordan Valley to the hubs of the Kingdom of Israel – Shechem and Samaria.
- The third type of fort includes three sites: The rectangular-rounded fort at el-Makhruq (2) (Site 73), the fort at 'Ein Abu Daraj (Site 154), and the possible fort at Umm Hallal (Site 156). The first one is positioned on a hill on the Makhruq ridge, close to the Jericho–Beit She'an road. Its outline is prominent and the supporting walls and the fortification wall are preserved. 'Ein Abu Daraj is a fortified tower with courtyards situated on the narrow ridge south of the spring. This positioning protected the spring and the approaches to it, and possibly the north-western approaches to the Sartaba range. The site at Umm Hallal has been almost totally destroyed, and the assumption that it is a fort is based on the remaining masonry and its dominant location. The number of forts in the Jordan valley again underlines the geo-political and military function of the region.

The commonest type of site is the farm. We recognized the IA farm as an architectonic unit with a dwelling structure or structures and an adjacent courtyard. The family farm exploits the land at the fringes of the desert for growing annual crops – wheat and barley. Eitam (1996: 695–699) also found and published quarried installations which indicate horticulture: oil and wine. Most of the farms were established on virgin soil during this period, and their



23. Map of the Iron Age II Period sites.

main distribution was along the Jordan Valley strip and north of the Sartaba.

Another common type is the enclosure – which was widespread in the desert fringes. Most of the enclosures are circular, indicating that they served as foundations of animal pens (but the issue has not been checked, and the definition is based on its resemblance to a modern animal pen). Most of the ‘foot-shaped’ enclosures continued from the IA I to the IA II.

Many sherd scatters have also been recorded, and it is likely that some of these sites were nomad tent encampments. This assumption is based on the absence of architecture, which under the conditions of the desert fringes would have been at least partly preserved.

Many cave sites have been recorded along the narrow Jordan Valley, and especially in the low-lying cliffs descending to the valley floor. These sites served as dwellings and for the upkeep of herds alike. The caves are generally not deep because of the soft conglomerate rock. In front of the caves there are almost always built courtyards. The sooty ceilings are proof of habitation in the caves, but it is difficult to determine if it was a matter of continuous residence or temporary presence of shepherds. The pottery scattered in front of the caves is evidence of continuous human activity over thousands of years.

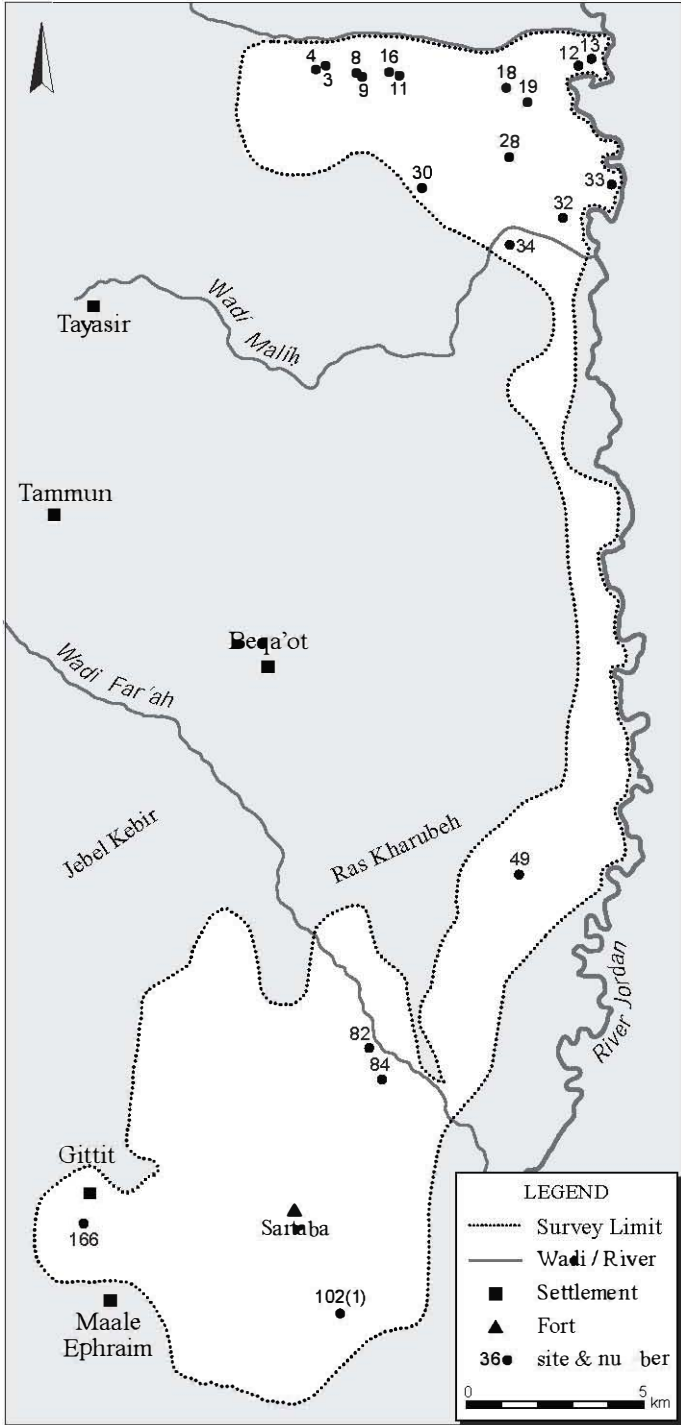
8. The Iron Age III (Assyrian and Neo-Babylonian Period, 722-535 BCE) and Persian Period (535-332 BCE)

There is a gap in our data about the Jordan Valley during the 200 years between the Assyrian and Persian conquests. To the period of the Assyrian rule, which we designated IA III (722-586 BCE), it is necessary to add the interlude of the short Neo-Babylonian rule (Lipschits and Blenkinsopp 2003). Almost no IA III sites have been found, although it would be surprising if the Assyrians failed to show interest in the Jordan Valley. Perhaps future research will add to our knowledge about these periods.

In the Persian period there was a notable decline of habitation along the Jordan Valley and the Sartaba range, indicated by the 20 sites discovered. This decline can also be seen in the desert fringes (Zertal 2008: 90-92).

Elsewhere (Zertal 1999: 78-79) we have defined the Persian period as having a pronounced ‘Mediterranean’ character from the political and economic aspects. The large settlement concentration in the Mediterranean hill country is prominent, and specifically so around the capital Samaria and the Dothan Valley. We have suggested that the period's economy relied mainly on the production of oil and wine for export: according to this concept, the desert fringes served mostly as marginal land. The population also was not attracted to this territory, and most of the IA sites were deserted.

We found evidence of habitation in the Persian period only in the south of



24. Map of the Persian Period sites.

Beit She'an Valley, mainly in the small tells already mentioned. The rest of the valley was an almost completely neglected.

9. Hellenistic Period (332-63 BCE)

There was a notable revival of settlement in the region during this period. The 34 sites in which Hellenistic sherds have been found are divided into two main groups.

The first is the southern Beit She'an Valley (12 sites). The settlement here was fairly dense and stable, and Hellenistic pottery was found mainly in small tells in the region. Only one site (el-Marma, Site 34), was founded in the Persian or Hellenistic period.

The second group, 18 sites surrounding the Sartaba range, is more interesting. The fortress of Alexandrion was founded by Alexander Jannaeus during this period. It is likely that the small sites surrounding the fortress served as its agricultural hinterland. Almost all the sites containing Hellenistic pottery were founded earlier, mostly during the IA II (after a long spell with no Persian presence).

The other areas discussed in this volume continued to be almost deserted (only three sites in the Jordan Valley and one in Nahal Tirzah).

Generally, and despite the recovery, the region in the Hellenistic period was quite poor. The historical data discussed below suggests that the Jordan Valley was on the line of march and battle of the various Hellenistic armies, with no significant settlement.

10. The Roman Periods: the Early (63 BCE-73 CE) and the Late (73-313 CE)

The number of settlements in the ER (Herodian) period marks the beginning of a change after the Hellenistic period. Most of the 46 settlements are situated on earlier sites. The number of sites around the Sartaba (22) raises the question of the nature of the Herodian fortress of Alexandrion in its relation to its probable hinterland. The question is whether the Alexandrion fortress was isolated from its vicinity, or did it enjoy a hinterland – agricultural, security or otherwise? The erection of the fortress by Alexander Jannaeus at the end of the 1st century BCE was probably not accompanied by dense settlement of the vicinity. The population peaked in the LR period, and the relationship of this to the fortress of Alexandrion is still unclear. During the LR period, from the end of the 1st century CE and onward, there was an unprecedented influx to the district, especially in the hills north of the Sartaba range.

Following is a comparison of the two periods: